



JUL – Angola's National Logistics Single Window (Phase 1)

Integration Control Document

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CNCA Certificate Issuance Process

Amendment and Cancellation Capabilities

Version 3.0
Abu Dhabi Ports
November 14, 2025

Document Control

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Key Stakeholders

Abu Dhabi Ports - JUL System

Role	Name	Email
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ARCCLA - SINTECE System

Role	Name	Email
Integration Lead	TBD	
Technical Support	Support Team	
Project Manager	TBD	

Version History

Version	Date	Author	Description of Changes
1.0	2025-11-12	Linoy Pappachan Malakkaran	Initial ICD creation with technical specifications for JUL-SINTECE integration. Defined API specifications, data models, integration workflows, security requirements, and testing procedures.
2.0	2025-11-13	Linoy Pappachan Malakkaran	Added amendment and cancellation APIs, approval workflows, status management with canAmend/canCancel flags, validation framework, and error handling. Added process flow improvements and implementation recommendations.
3.0	2025-11-14	Linoy Pappachan Malakkaran	Removed flow diagrams, webhook references, payment processing references. Added Keycloak SSO authentication, NIF validation API, invoice download API, status retrieval API, and consolidated CNCA certificate submission API. Updated table format to 6 columns.
3.0-U	2025-11-17	Linoy Pappachan Malakkaran	Split Format/Derivation Logic columns, added WCO Data Model format standards to all API tables.

1. Introduction

1.1 Purpose

This Interface Control Document (ICD) defines the technical specifications and interface requirements for integrating the JUL system (Abu Dhabi Ports) with the SINTECE system (ARCCLA) for CNCA certificate issuance.

Key Features: - Certificate submission and approval workflow - Amendment and cancellation requests - Master data synchronization - Status tracking and notifications

1.2 Scope

In Scope: - Certificate request lifecycle (creation to issuance) - Amendment and cancellation workflows - API specifications and data models - Authentication and authorization - Validation and error handling

Out of Scope: - Internal SINTECE business logic - Payment gateway implementation - Mobile application development - Legacy system migration

1.3 Audience

- **Developers:** API implementation and integration
- **Architects:** System design and technical decisions
- **QA Engineers:** Testing and validation
- **Business Analysts:** Requirements validation
- **Project Managers:** Implementation tracking

1.4 Key Terms

Term	Definition
ARCCLA	Angolan regulatory agency responsible for cargo certification
BL	Bill of Lading - shipping document required for certificates
CNCA	Certificado Nacional de Carga de Angola - required certificate for Angola cargo
CTN	Certificate/Cargo Tracking Note - internal certificate reference
DUP	Declaration of Unique Property - unique customs identifier
JUL	Abu Dhabi Ports system for CNCA certificate management

Term	Definition
SINTECE	ARCCLA system for certificate processing and approval
Amendment	Request to modify submitted certificate data
Cancellation	Request to cancel submitted certificate
canAmend/canCancel	System flags indicating eligibility for amendment/cancellation

2. System Overview

2.1 System Architecture

JUL System (Abu Dhabi Ports) - Trader portal for CNCA certificate management - User interface for request creation and document upload
 - Amendment/cancellation request capabilities - Certificate download and status tracking
 - Real-time status tracking and notifications

SINTECE System (ARCCLA) - Backend certificate processing and approval system - ARCCLA broker review workflow - Amendment/cancellation approval processes - Certificate issuance and status management - Master data management and reporting

Integration Layer - RESTful APIs with JSON data format - JWT authentication and secure communication - Real-time API calls and status notifications - Error handling and audit logging

2.2 Integration Pattern

Communication: - RESTful APIs with standard HTTP methods (GET, POST, PUT, DELETE) - JSON data format with schema validation - HTTPS/TLS encryption and JWT authentication - OData query support for filtering and pagination

Data Flow:

- Initial CTN Creation:** JUL creates basic CTN record with core information via POST /api/ctns
- Section-wise Data Addition:** JUL adds detailed information in separate steps (internal to JUL system):
 - **Addresses:** Shipper, consignee, forwarder, notify party details
 - **Goods:** Cargo descriptions, classifications, weights, values
 - **Containers:** Container types, numbers, seals
 - **Tracking:** Transport routes, vessels, ports, schedules
 - **Attachments:** Supporting documents

3. **Certificate Validation:** SINTECE validates complete certificate data with business rules
 4. **Visa Request Submission:** JUL submits for approval via POST `/api/ctns/actions/requestvisa/{id}`
 5. **ARCCLA Review:** ARCCLA broker reviews and approves/rejects certificate
 6. **Certificate Issuance:** Certificate issued with amendment/cancellation capabilities
 7. **Status Updates:** Status communicated via API polling and notifications
-

Amendment Flow:

1. JUL validates amendment eligibility using canAmend status flag
2. JUL submits amendment request via POST `/api/CTNs/{id}/amendments` with change tracking
3. SINTECE validates amendment request and notifies ARCCLA broker
4. ARCCLA broker reviews amendment in SINTECE with impact analysis
5. SINTECE sends amendment approval/rejection to JUL via notifications
6. If approved, SINTECE updates certificate and certificate status
7. JUL updates certificate status and notifies stakeholders

Cancellation Flow:

1. JUL validates cancellation eligibility using canCancel status flag
2. JUL submits cancellation request via POST `/api/CTNs/{id}/cancellations` with reason documentation
3. SINTECE validates cancellation request and calculates financial impact
4. ARCCLA broker reviews cancellation in SINTECE with impact analysis
5. SINTECE sends cancellation approval/rejection to JUL via notifications
6. If approved, SINTECE processes cancellation and updates certificate status

2.3 System Actors

The following actors interact with the integrated system:

Actor	Role	System Access
Trader (Importer/Exporter)	Business entity shipping goods to/from Angola	JUL System (Web Portal)
Customs Broker / Freight Forwarder	Licensed agent representing trader with amendment/cancellation capabilities	JUL System (Web Portal with Amendment/Cancellation Features)
ARCCLA Broker	Government official authorized to approve certificates, amendments, and cancellations	SINTECE System (Internal Portal with Amendment/Cancellation Workflows)

Enhanced User Interaction Flow:

Standard Certificate Flow:

1. **Trader** logs into enhanced JUL → Creates certificate request → Uploads documents → Nominates broker
2. **Customs Broker** logs into enhanced JUL → Accepts nomination → Completes application with real-time validation → Submits to SINTECE
3. **ARCCLA Broker** logs into enhanced SINTECE → Reviews request with decision support tools → Approves/Rejects → Generates certificate (if approved)
4. **Trader** and **Customs Broker** receive certificate in JUL → Download and use for customs clearance

Amendment Flow:

1. **Customs Broker** checks amendment eligibility in JUL → Creates amendment request → Documents changes
2. **ARCCLA Broker** receives amendment notification in SINTECE → Reviews changes with impact analysis → Approves/Rejects

3. **Customs Broker** receives amendment decision in JUL → Updates stakeholders
4. **Trader** receives updated certificate with amendment history in JUL

Cancellation Flow:

1. **Customs Broker** checks cancellation eligibility in JUL → Creates cancellation request → Documents reasons
2. **ARCCLA Broker** receives cancellation notification in SINTECE → Reviews with impact analysis → Approves/Rejects
3. **Customs Broker** receives cancellation decision in JUL → Updates stakeholders
4. **Trader** receives cancellation confirmation

2.4 Amendment and Cancellation Capabilities

Amendment Management: - Real-time eligibility validation based on certificate status - Change tracking with before/after value comparison - Impact analysis for proposed amendments - Role-based approval workflow - Complete audit trail with history

Cancellation Management: - Real-time eligibility validation with financial impact assessment - Reason code documentation - Role-based approval workflow with escalation - Complete audit trail

Enhanced Status Management: - Dynamic eligibility flags (canAmend, canCancel) based on real-time business rules - Enhanced status workflow supporting amendment/cancellation states - Real-time status tracking and notifications - Comprehensive status history with audit trail - Status-based UI enhancements for improved user experience

Advanced Validation Framework: - Multi-level validation (field-level, business rule, cross-system) - Real-time validation with immediate feedback - Configurable validation rules with role-based exceptions - Comprehensive error messaging with corrective action guidance - Integration testing validation for amendment/cancellation workflows

3. Process Flows

3.1 As-Is Process Analysis

The current SINTECE workflow consists of 15 distinct steps (LC-AR-CNCA-01 through LC-AR-CNCA-15) that involve multiple manual handoffs between traders, customs brokers, and ARCCLA staff.

 **Process Inefficiencies:** - Multiple email-based communications causing delays - Manual document handling and verification processes - Limited real-

time visibility for stakeholders - Duplicate data entry requirements across systems - No automated amendment or cancellation capabilities

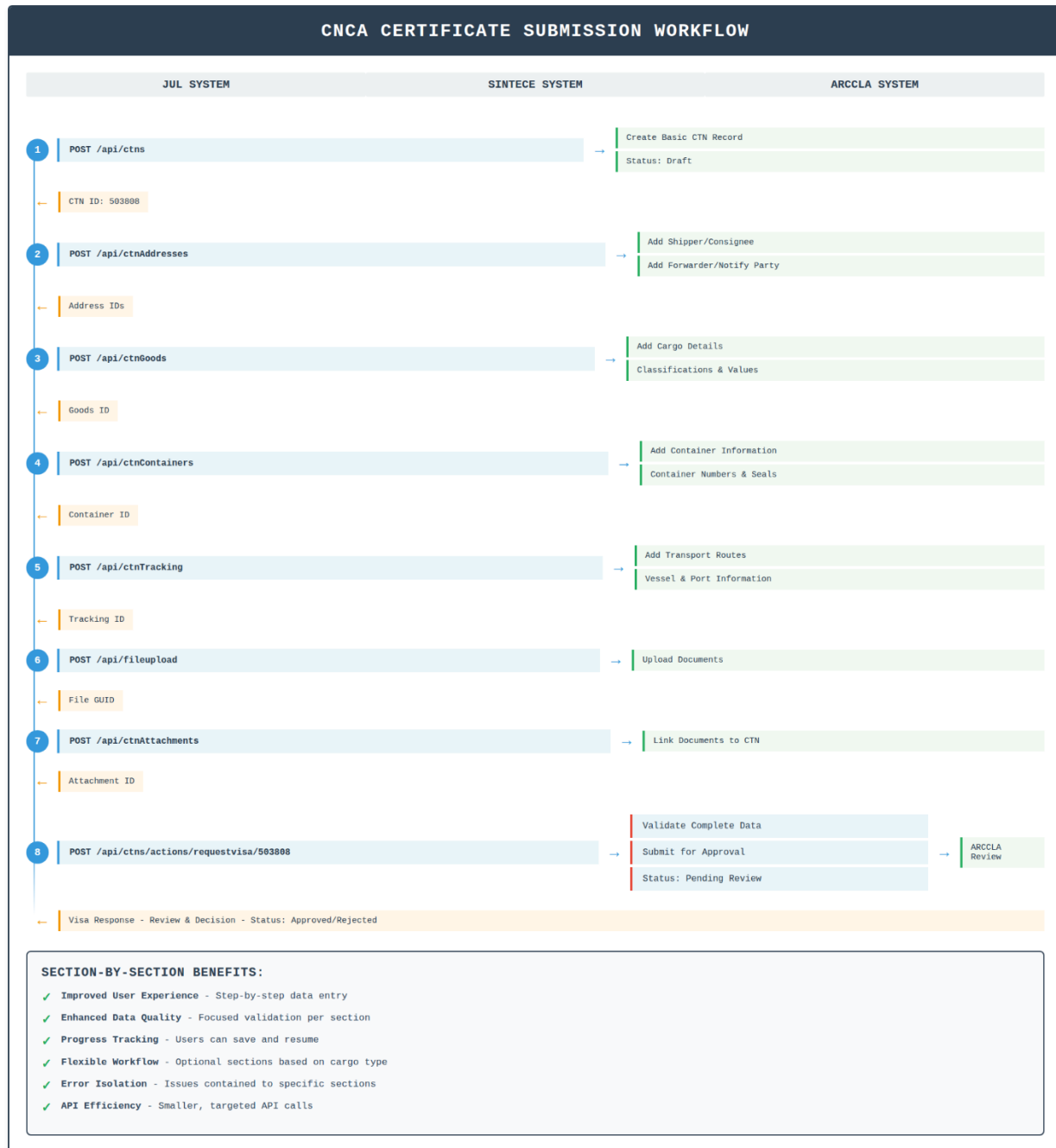
⚠ **Key Bottlenecks:** - ARCCLA broker manual review and approval steps - Payment processing and reconciliation delays
- Physical document submission requirements - Manual invoice generation and distribution

3.2 To-Be Process

The JUL-SINTECE integration introduces a fully digital workflow with section-by-section certificate submission capabilities to improve data accuracy and user experience.



3.2.1 Section-by-Section Certificate Submission Workflow



✓ Core Certificate Process Improvements:

1. **Digital Initiation:** Traders initiate requests directly through JUL system with section-by-section data entry
2. **Progressive Document Building:** Step-by-step completion of certificate sections for improved data quality

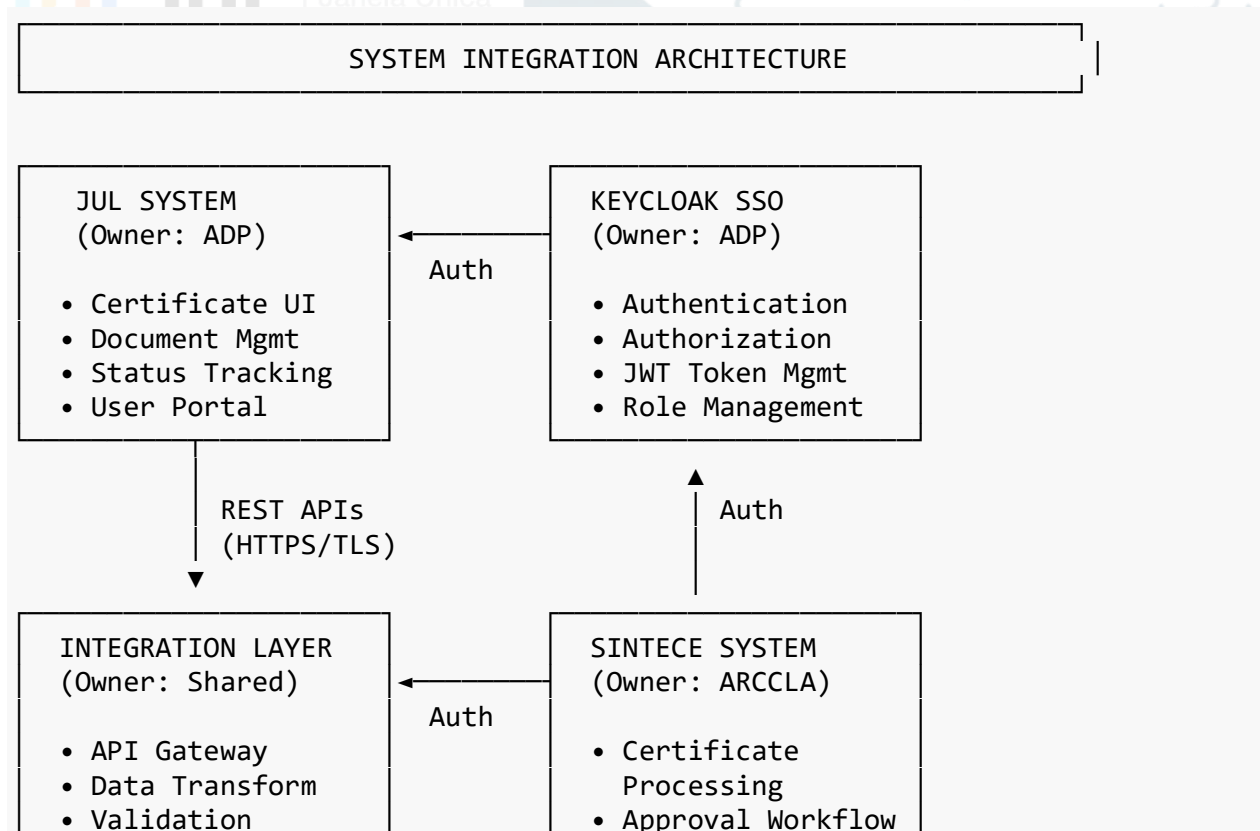
3. **Automated Validation:** Real-time validation of submitted data at each section
4. **Streamlined Approval:** ARCCLA brokers receive complete, validated requests for review
5. **Integrated Payment:** Seamless payment processing with automatic reconciliation
6. **Digital Certificate Issuance:** Electronic certificate generation and delivery
- 🚀 **Enhanced Capabilities Integration:** - Real-time amendment requests with ARCCLA approval workflow - Cancellation management with refund processing capabilities - Status polling with eligibility flags for allowed operations - Comprehensive audit trail and notifications system

Benefits of Enhanced Process: - Fully digital workflow with no physical documents - Real-time status visibility for all parties - Integrated amendment and cancellation capabilities - Automated payment processing - Single data entry point - Complete audit trail

3.3 System Integration Architecture

The integration establishes comprehensive touchpoints between JUL and SINTECE systems with a modern API-driven architecture and centralized authentication.

System Integration Overview:



- Error Handling

- Master Data Mgmt
- Status Management

KEY INTEGRATION POINTS:

- Authentication: Keycloak SSO with JWT tokens
- Master Data Sync: Real-time via REST APIs
- Certificate Submission: Section-by-section APIs
- Status Updates: Polling-based status retrieval
- Document Exchange: Secure file transfer APIs

Component Ownership Table:

Component	Owner	Responsibilities
JUL System	Abu Dhabi Ports (ADP)	User interface, certificate request initiation, document upload, status display, user notifications, certificate download
Keycloak SSO	Abu Dhabi Ports (ADP)	User authentication, authorization, JWT token issuance and validation, session management, role-based access control
Integration Layer	Shared (ADP + ARCCLA)	API gateway, request/response transformation, data validation, error handling, logging and monitoring
SINTECE System	ARCCLA (Angola)	Certificate processing, approval workflows, master data management, certificate issuance, status management

Integration Flow Description:

1. User authenticates via Keycloak SSO (OpenID Connect/OAuth2)
2. JUL obtains JWT access token for API authorization
3. JUL sends certificate data to SINTECE via REST APIs
4. Integration layer validates and transforms requests
5. SINTECE processes certificate and updates status
6. JUL polls SINTECE for status updates
7. Certificate issued and available for download

Data Synchronization Points: - Master data synchronization (countries, ports, cargo types) - User account and role management coordination - Certificate status and lifecycle management - Amendment and cancellation request handling

API Integration Points: - Authentication and authorization endpoints - Certificate submission and retrieval APIs
- Amendment request creation and approval APIs - Cancellation request creation and approval APIs - Status monitoring and notification APIs

Business Process Touchpoints: - ARCCLA broker approval workflows - Document management and storage - Audit logging and compliance reporting

3.4 Amendment Workflow Process

The amendment workflow provides controlled modification capabilities for issued certificates with comprehensive ARCCLA approval processes.

Amendment Request Initiation:

1. **Eligibility Check:** System validates certificate status and amendment permissions
2. **Request Creation:** Customs broker selects fields to amend and provides change reasons
3. **Validation:** System performs business rule validation on proposed changes
4. **Submission:** Amendment request is submitted to ARCCLA for review

ARCCLA Review and Approval:

1. **Notification:** ARCCLA broker receives amendment request notification
2. **Impact Analysis:** Review of proposed changes and potential impacts
3. **Decision:** Approval or rejection with detailed reasoning
4. **Processing:** For approved amendments, certificate is updated and new invoice generated

Amendment Eligibility Rules:

- Certificate status must be “Submitted”, “Approved”, or “Issued”
- No pending amendments or cancellations exist
- Certificate not yet used for customs clearance
- Amendment requested within allowed timeframe (typically 24 hours after issuance)
- User has appropriate amendment permissions

3.5 Cancellation Workflow Process

The cancellation workflow provides controlled termination of certificate requests with comprehensive financial impact management and refund processing.

Cancellation Request Initiation:

1. **Eligibility Check:** System validates certificate status and cancellation permissions
2. **Impact Assessment:** The System calculates impact and eligibility
3. **Request Creation:** The Customs broker provides the cancellation reason and additional details
4. **Submission:** Cancellation request is submitted to ARCCLA for review

ARCCLA Review and Processing:

1. **Notification:** ARCCLA broker receives cancellation request notification
2. **Review:** Validation of cancellation reason and business justification
3. **Analysis:** Review of cancellation impact and timing
4. **Decision:** Approval or rejection of cancellation request
5. **Processing:** For approved cancellations, the certificate is cancelled

Cancellation Eligibility Rules:

- Certificate status allows cancellation (not yet issued or used for customs clearance)
- Cancellation requested within the allowed timeframe
- No active amendments in progress - Valid business reason for cancellation provided

3.6 ARCCLA Approval Workflows

The ARCCLA approval workflows are enhanced to support the new amendment and cancellation capabilities:

Enhanced Review Dashboard:

- Unified queue for certificate submissions, amendments, and cancellations
- Priority-based processing with configurable business rules
- Rich context information for informed decision making
- Integrated communication tools for broker-to-broker coordination

Approval Decision Framework:

- Structured decision criteria for different request types
- Automated recommendations based on historical patterns
- Escalation procedures for complex or high-value requests
- Audit trail capture for all approval decisions

Workflow Automation:

- Automatic assignment of requests to available brokers
- SLA monitoring with escalation triggers
- Batch processing capabilities for high-volume periods
- Integration with existing SINTECE approval processes

Enhanced Notification System:

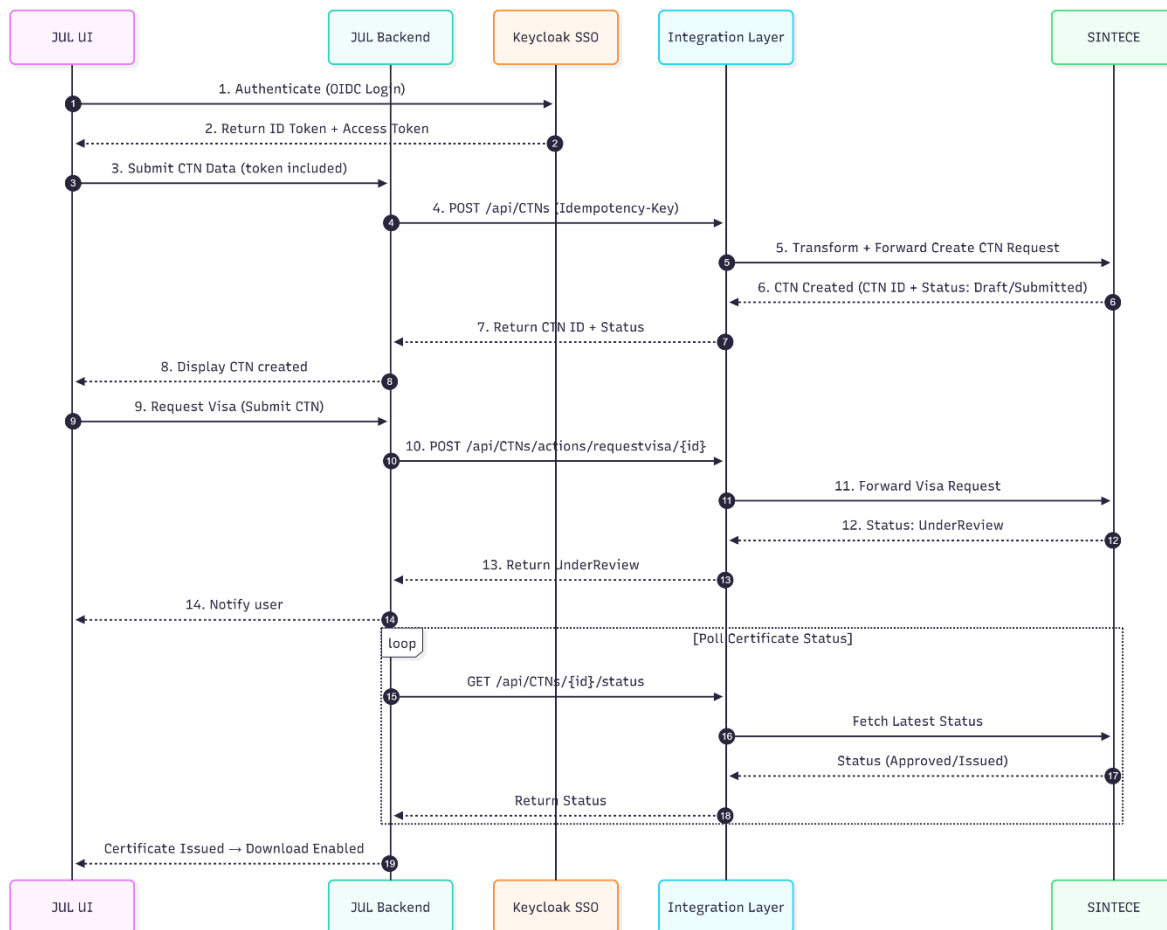
- Real-time notifications to all stakeholders
- Email and system notifications with rich content
- Configurable notification preferences by user role

3.7 Sequence Diagrams

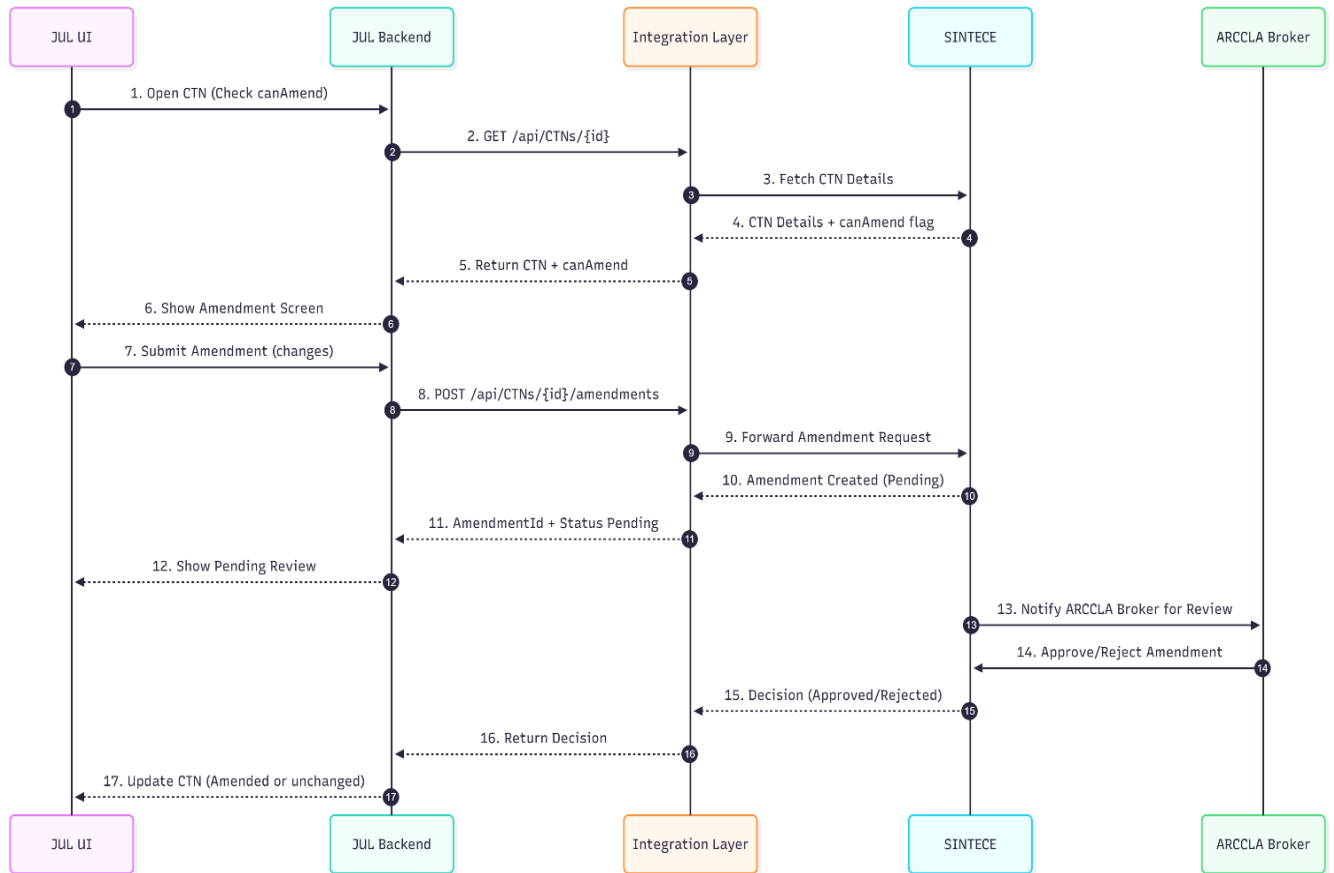
3.7 Sequence Diagrams

This section describes logical sequence flows between JUL, Keycloak, Integration Layer, and SINTECE.

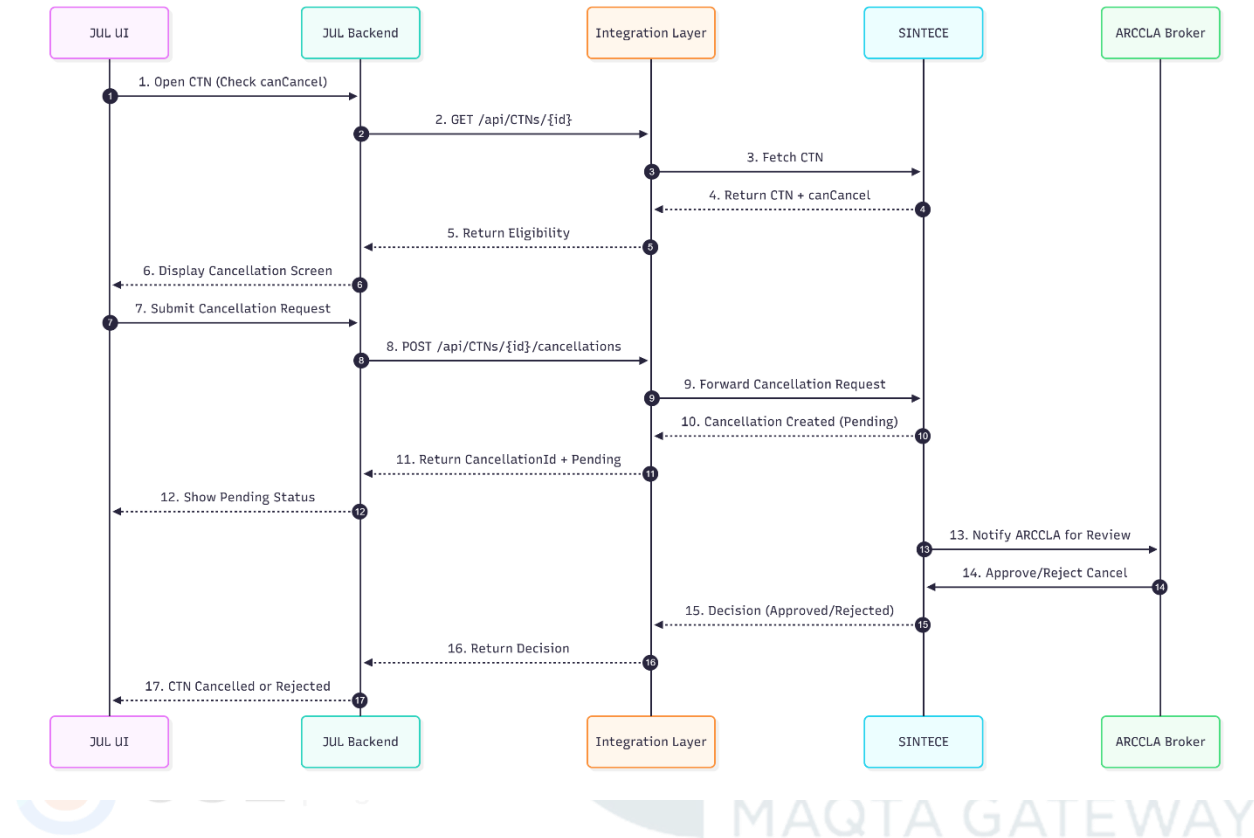
3.7.1 CTN Creation and Visa Request Sequence



3.7.2 Amendment Request Sequence



3.7.3 Cancellation Request Sequence



4. Data Models

4.1 Core CTN Entity Structure

The Certificate Tracking Note (CTN) entity serves as the foundation for all certificate operations. The enhanced data model includes additional fields to support amendment and cancellation workflows:

```

{
  "id": "string (UUID)",
  "certificateNumber": "string (auto-generated)",
  "requestDate": "datetime (ISO 8601)",
  "submissionDate": "datetime (ISO 8601)",
  "status": "string (enum)",
  "canAmend": "boolean",
  "canCancel": "boolean",
  "lastAmendmentDate": "datetime (ISO 8601, nullable)",
  "lastCancellationAttempt": "datetime (ISO 8601, nullable)",
  "isAmended": "boolean (default: false)",
}

```

```
"isCancelled": "boolean (default: false)",
"originalCertificateId": "string (UUID, nullable)",
"trader": {
  "traderId": "string (UUID)",
  "traderName": "string (required, max 200)",
  "traderEmail": "string (required, valid email)",
  "traderPhone": "string (required)",
  "traderAddress": {
    "street": "string (required, max 200)",
    "city": "string (required, max 100)",
    "country": "string (required, ISO country code)",
    "postalCode": "string (optional, max 20)"
  }
},
"customsBroker": {
  "brokerId": "string (UUID)",
  "brokerName": "string (required, max 200)",
  "brokerEmail": "string (required, valid email)",
  "brokerLicenseNumber": "string (required, max 50)",
  "brokerCompany": "string (required, max 200)",
  "isNominated": "boolean (default: false)",
  "nominationDate": "datetime (ISO 8601, nullable)"
},
"shipment": {
  "billOfLadingNumber": "string (required, max 50)",
  "dupNumber": "string (required, max 50)",
  "vesselName": "string (required, max 200)",
  "voyageNumber": "string (required, max 50)",
  "portOfLoading": "string (required, port code)",
  "portOfDischarge": "string (required, port code)",
  "estimatedTimeOfArrival": "datetime (ISO 8601)",
  "actualTimeOfArrival": "datetime (ISO 8601, nullable)"
},
"goods": [
  {
    "itemNumber": "integer (sequence)",
    "description": "string (required, max 500)",
    "hsCode": "string (required, HS code format)",
    "quantity": "number (required, positive)",
    "unit": "string (required, standard units)",
    "weight": "number (required, positive, kg)",
    "volume": "number (optional, positive, m³)",
    "value": "number (required, positive, USD)",
    "origin": "string (required, ISO country code)",
    "manufacturer": "string (optional, max 200)"
  }
],
"containers": [
  {
```

```

    "containerNumber": "string (required, max 20)",
    "containerType": "string (required, ISO container type)",
    "sealNumbers": ["string (max 50)"],
    "tareWeight": "number (required, positive, kg)",
    "grossWeight": "number (required, positive, kg)",
    "cargoWeight": "number (required, positive, kg)"
  },
],
"payment": {
  "invoiceNumber": "string (auto-generated)",
  "amount": "number (required, positive, USD)",
  "currency": "string (required, ISO currency code)",
  "paymentStatus": "string (enum)",
  "paymentDate": "datetime (ISO 8601, nullable)",
  "paymentMethod": "string (enum)",
  "refundAmount": "number (optional, positive, USD)",
  "refundStatus": "string (enum, nullable)",
  "refundDate": "datetime (ISO 8601, nullable)"
},
"audit": {
  "createdBy": "string (UUID)",
  "createdDate": "datetime (ISO 8601)",
  "lastModifiedBy": "string (UUID)",
  "lastModifiedDate": "datetime (ISO 8601)",
  "version": "integer (auto-increment)"
}
}

```

Key Enhancements:

- Added canAmend and canCancel eligibility flags
- Enhanced payment model with refund capabilities
- Added amendment and cancellation tracking fields
- Comprehensive audit trail with versioning

4.2 Amendment Entity Structure

The amendment entity captures all modification requests with complete change tracking:

```

{
  "amendmentId": "string (UUID)",
  "originalCertificateId": "string (UUID, foreign key)",
  "amendmentNumber": "string (auto-generated, sequential)",
  "requestDate": "datetime (ISO 8601)",
  "submittedBy": "string (UUID, foreign key)",
  "submittedByRole": "string (enum: CustomsBroker, Trader)",
  "status": "string (enum: Pending, UnderReview, Approved, Rejected, Cancelled)",
  "requestReason": "string (required, max 500)",
  "businessJustification": "string (optional, max 1000)",

```

```
"requestedChanges": [
  {
    "fieldPath": "string (required, JSON path notation)",
    "fieldName": "string (required, human-readable)",
    "oldValue": "any (original value)",
    "newValue": "any (proposed new value)",
    "changeReason": "string (required, max 200)",
    "validationStatus": "string (enum: Valid, Invalid, Warning)"
  }
],
"arcclaReview": {
  "assignedTo": "string (UUID, nullable)",
  "assignedDate": "datetime (ISO 8601, nullable)",
  "reviewStarted": "datetime (ISO 8601, nullable)",
  "reviewCompleted": "datetime (ISO 8601, nullable)",
  "decision": "string (enum: Approved, Rejected, nullable)",
  "decisionReason": "string (optional, max 1000)",
  "reviewComments": "string (optional, max 2000)",
  "impactAssessment": {
    "riskLevel": "string (enum: Low, Medium, High)",
    "affectedSystems": ["string"],
    "complianceImpact": "string (optional, max 500)",
    "financialImpact": "number (optional, USD)"
  }
},
"fees": {
  "amendmentFee": "number (required, positive, USD)",
  "currency": "string (required, ISO currency code)",
  "calculationMethod": "string (required)",
  "invoiceNumber": "string (auto-generated, nullable)",
  "paymentStatus": "string (enum)",
  "paymentDate": "datetime (ISO 8601, nullable)"
},
"processing": {
  "applicationStarted": "datetime (ISO 8601, nullable)",
  "applicationCompleted": "datetime (ISO 8601, nullable)",
  "certificateUpdated": "datetime (ISO 8601, nullable)",
  "notificationsSent": "datetime (ISO 8601, nullable)",
  "processingErrors": [
    {
      "errorCode": "string",
      "errorMessage": "string",
      "errorDate": "datetime (ISO 8601)"
    }
  ]
},
"audit": {
  "createdBy": "string (UUID)",
  "createdDate": "datetime (ISO 8601)",
```

```

    "lastModifiedBy": "string (UUID)",
    "lastModifiedDate": "datetime (ISO 8601)",
    "workflowHistory": [
      {
        "action": "string",
        "performedBy": "string (UUID)",
        "performedDate": "datetime (ISO 8601)",
        "comments": "string (optional)",
        "systemGenerated": "boolean"
      }
    ]
  }
}

```

4.3 Cancellation Entity Structure

The cancellation entity manages certificate termination requests with comprehensive financial tracking:

```

{
  "cancellationId": "string (UUID)",
  "originalCertificateId": "string (UUID, foreign key)",
  "cancellationNumber": "string (auto-generated, sequential)",
  "requestDate": "datetime (ISO 8601)",
  "submittedBy": "string (UUID, foreign key)",
  "submittedByRole": "string (enum: CustomsBroker, Trader)",
  "status": "string (enum: Pending, UnderReview, Approved, Rejected, Cancelled)",
  "cancellationReason": {
    "reasonCode": "string (enum: required)",
    "reasonDescription": "string (required, max 500)",
    "additionalDetails": "string (optional, max 1000)",
    "urgencyLevel": "string (enum: Normal, High, Critical)"
  },
  "financialImpact": {
    "originalAmount": "number (required, USD)",
    "refundEligibility": "boolean",
    "refundAmount": "number (optional, USD)",
    "cancellationFee": "number (optional, USD)",
    "netRefund": "number (optional, USD)",
    "refundCalculationMethod": "string (required)",
    "refundJustification": "string (optional, max 500)"
  },
  "arcclaReview": {
    "assignedTo": "string (UUID, nullable)",
    "assignedDate": "datetime (ISO 8601, nullable)",
    "reviewStarted": "datetime (ISO 8601, nullable)",
    "reviewCompleted": "datetime (ISO 8601, nullable)",
    "decision": "string (enum: Approved, Rejected, nullable)"
  }
}

```

```

"decisionReason": "string (optional, max 1000)",
"reviewComments": "string (optional, max 2000)",
"riskAssessment": {
  "fraudRisk": "string (enum: Low, Medium, High)",
  "complianceRisk": "string (enum: Low, Medium, High)",
  "operationalRisk": "string (enum: Low, Medium, High)",
  "overallRisk": "string (enum: Low, Medium, High)"
},
"refundProcessing": {
  "refundInitiated": "datetime (ISO 8601, nullable)",
  "refundCompleted": "datetime (ISO 8601, nullable)",
  "refundMethod": "string (enum: nullable)",
  "refundReference": "string (nullable)",
  "refundStatus": "string (enum: nullable)",
  "processingErrors": [
    {
      "errorCode": "string",
      "errorMessage": "string",
      "errorDate": "datetime (ISO 8601)"
    }
  ]
},
"audit": {
  "createdBy": "string (UUID)",
  "createdDate": "datetime (ISO 8601)",
  "lastModifiedBy": "string (UUID)",
  "lastModifiedDate": "datetime (ISO 8601)",
  "workflowHistory": [
    {
      "action": "string",
      "performedBy": "string (UUID)",
      "performedDate": "datetime (ISO 8601)",
      "comments": "string (optional)",
      "systemGenerated": "boolean"
    }
  ]
}
}

```

4.4 Enhanced Status Management

The enhanced status management system provides comprehensive lifecycle tracking with dynamic eligibility determination:

Certificate Status Enumeration:

```

{
  "certificateStatuses": [
    "Draft",

```



```
"Submitted",  
"UnderReview",  
"RequiresAmendment",  
"Approved",  
"PaymentPending",  
"PaymentReceived",  
"Issued",  
"Active",  
"Amended",  
"Cancelled",  
"Expired",  
"Rejected"  
]  
}
```

Amendment Status Enumeration:

```
{  
  "amendmentStatuses": [  
    "Pending",  
    "UnderReview",  
    "RequiresInformation",  
    "Approved",  
    "Rejected",  
    "PaymentPending",  
    "PaymentReceived",  
    "Processing",  
    "Completed",  
    "Cancelled",  
    "Expired"  
  ]  
}
```

Cancellation Status Enumeration:

```
{  
  "cancellationStatuses": [  
    "Pending",  
    "UnderReview",  
    "RequiresInformation",  
    "Approved",  
    "Rejected",  
    "RefundPending",  
    "RefundProcessing",  
    "RefundCompleted",  
    "Completed",  
    "Cancelled"  
  ]  
}
```

Eligibility Rules Matrix: The system dynamically calculates eligibility flags based on the following business rules:

Amendment Eligibility (canAmend):

- Certificate status is in: ["Submitted", "Approved", "Issued", "Active"]
- No pending amendments exist for the certificate
- No pending cancellations exist for the certificate
- Certificate was issued within the allowed amendment timeframe (configurable)
- User has appropriate permissions (CustomsBroker or authorized Trader)
- Certificate has not been used for customs clearance

Cancellation Eligibility (canCancel):

- Certificate status is in: ["Submitted", "Approved", "PaymentPending", "PaymentReceived", "Issued"]
- No pending amendments exist for the certificate
- No pending cancellations exist for the certificate
- Certificate was submitted within the allowed cancellation timeframe (configurable)
- User has appropriate permissions (CustomsBroker or authorized Trader)
- Certificate has not been used for customs clearance

Dynamic Status Calculation:

The system recalculates eligibility flags whenever:

- Certificate status changes
- Amendment or cancellation requests are submitted
- Time-based rules expire
- User permissions are modified
- Business rules are updated by administrators

5. Core API Specifications

5.1 Master Data APIs

Master data APIs provide essential reference data required for certificate creation, validation, and business operations.

5.1.1 Cargo Types API

Business Purpose: Manages cargo classification data essential for determining certificate fees, validation rules, and compliance requirements. Critical for proper cargo categorization in international shipping, affecting customs documentation, handling procedures, insurance requirements, and regulatory compliance.

Endpoint: GET /api/CargoTypes

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	\$sort	String	an..100	O	OData sort parameter	CargoType_Desc
2	CargoType_Desc	String	an..200	O	Filter by cargo type description	CONTAINER
3	active	Boolean	n1	O	Filter active records only	1

UI Validation Rules:

- CARGO_UI_001: CargoType_Desc filter optional, 1-200 characters
- CARGO_UI_002: Sort parameter from predefined fields only
- CARGO_UI_003: Max 100 records per page

Business Validation Rules:

- CARGO_BV_001: Only active cargo types can be used for new CTN creation
- CARGO_BV_002: Container cargo requires container details
- CARGO_BV_003: Dangerous goods cargo requires special documentation

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	Id	Integer	n..10	M	Unique cargo type identifier	1
2	CargoType_Code	String	an..10	M	Cargo type code	CONT
3	CargoType_Desc	String	an..200	M	Cargo type description	CONTAINER
4	Active	Boolean	n1	M	Active status indicator	true
5	CreatedOn	DateTime	DATE	O	Record creation timestamp	2023-01-15T10:30:00Z

Error Codes:

- CARGO_E001: "Invalid parameters" (HTTP 400)
- CARGO_E002: "No data found" (HTTP 404)
- CARGO_E003: "Access denied" (HTTP 403)

Sample HTTP Request:

```
GET /api/CargoTypes?$sort=CargoType_Desc&CargoType_Desc=&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "MultiLingualDescription": {
      "Id": 2183,
      "Translations": [
        {
          "Id": 2187,
          "LanguageISO": "en",
          "MultiLingualTextId": 2183,
          "Text": "CONTAINER"
        },
        {
          "Id": 2188,
          "LanguageISO": "fr",
          "MultiLingualTextId": 2183,
          "Text": "CONTENEUR"
        },
        {
          "Id": 2304,
          "LanguageISO": "pt",
          "MultiLingualTextId": 2183,
          "Text": "Contentor"
        }
      ]
    },
    "Code": "CONTAINER",
    "CargoType_Desc": "CONTAINER",
    "CreatedOn": null,
    "CreatedById": null,
    "ModifiedOn": null,
    "ModifiedById": null
  },
  {
    "Id": 2,
```

```

"MultiLingualDescription": {
  "Id": 2184,
  "Translations": [
    {
      "Id": 2189,
      "LanguageISO": "en",
      "Text": "BULK"
    },
    {
      "Id": 2190,
      "LanguageISO": "fr",
      "Text": "VRAC"
    },
    {
      "Id": 2309,
      "LanguageISO": "pt",
      "Text": "Sacrias ou carga fracionada"
    }
  ]
},
"Code": "BULK",
"CargoType_Desc": "BULK",
"CreatedOn": null,
"CreatedById": null,
"ModifiedOn": null,
"ModifiedById": null
}
]

```

Sample JSON Response:

```

[
  {
    "Id": 1,
    "CargoType_Code": "CONT",
    "CargoType_Desc": "CONTAINER",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  },
  {
    "Id": 2,
    "CargoType_Code": "BULK",
    "CargoType_Desc": "BULK CARGO",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,

```

```
"ModifiedById": null
}
```

5.1.2 Incoterms API

Business Purpose: Provides International Commercial Terms (Incoterms) data crucial for determining responsibility boundaries, costs, and risks between buyers and sellers in international trade transactions. Affects shipping responsibilities, insurance requirements, customs clearance obligations, and liability allocation throughout the shipping process.

Endpoint: GET /api/Incoterms

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	\$sort	String	an..100	O	OData sort parameter	IncotermCode
2	Incoterm Code	String	a3	O	Filter by Incoterm code	CIF
3	active	Boolean	n1	O	Filter active records only	1

UI Validation Rules:

- INCO_UI_001: IncotermCode filter optional, exactly 3 uppercase letters
- INCO_UI_002: Display code and description together
- INCO_UI_003: Default to “Active Only” for new certificates

Business Validation Rules:

- INCO_BV_001: CIF/CIP terms require insurance documentation
- INCO_BV_002: Some Incoterms not applicable for certain cargo types
- INCO_BV_003: Incoterm selection affects customs valuation

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	Id	Integer	n..10	M	Unique Incoterm identifier	2
2	Incoterm Code	String	a3	M	Incoterm code	CIF

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
3	Incoterm Desc	String	an..200	M	Incoterm description	Cost, Insurance and Freight
4	Active	Boolean	n1	M	Active status indicator	true

Error Codes:

- INCO_E001: "Invalid format" (HTTP 400)
- INCO_E002: "Not applicable for cargo type" (HTTP 422)
- INCO_E003: "Inactive Incoterm" (HTTP 422)

Sample JSON Request:

```
GET /api/Incoterms?$sort=IncotermCode&IncotermCode=&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 6,
    "IncotermCode": "CFR",
    "MultiLingualDescription": {
      "Id": 2211,
      "Translations": [
        {
          "Id": 2238,
          "LanguageISO": "en",
          "MultiLingualTextId": 2211,
          "Text": "Cost and Freight"
        }
      ]
    },
    "DescriptiveFields": "Cost and Freight"
  },
  {
    "Active": true,
    "CreatedOn": null,
    "ModifiedOn": "2023-01-11T16:51:28.8309058",
    "ModifiedById": 1,
    "ModifiedByLogin": "TCNT"
  }
],
{
  "Id": 2,
  "IncotermCode": "CIF",
```

```

"MultiLingualDescription": {
  "Id": 2207,
  "Translations": [
    {
      "Id": 2234,
      "LanguageISO": "en",
      "Text": "Cost, Insurance, Freight"
    }
  ],
  "DescriptiveFields": "Cost, Insurance, Freight"
},
"Active": true,
"CreatedOn": null,
"ModifiedOn": null,
"ModifiedById": null
}
]

```

5.1.3 Countries API

Business Purpose: Essential for international trade compliance, determining applicable regulations, duties, restrictions, and routing requirements for cargo shipments. Critical for sanctions screening, trade agreement benefits, customs procedures, and ensuring compliance with import/export regulations between trading nations.

Endpoint: GET /api/Countries

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	\$sort	String	an..100	O	OData sort parameter	Country_Name
2	Country_Name	String	an..100	O	Filter by country name	Angola
3	Exporting Country	Boolean	n1	O	Filter exporting countries	true
4	active	Boolean	n1	O	Filter active records only	1

UI Validation Rules:

- COUNTRY_UI_001: Country name filter optional, 1-100 characters
- COUNTRY_UI_002: Auto-complete dropdown with country name and code
- COUNTRY_UI_003: Visual indicator for sanctioned/restricted countries

Business Validation Rules:

- COUNTRY_BV_001: Angola must be origin or destination for CNCA certificates
- COUNTRY_BV_002: Sanctioned countries require special authorization
- COUNTRY_BV_003: Country code must match ISO 3166-1 standard

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	Id	Integer	n..10	M	Unique country identifier	10
2	Country_Name	String	an..100	M	Country name	Angola
3	Country_Code	String	a2	M	ISO 3166 Alpha 3 country code	AO
4	Exporting Country	Boolean	n1	M	Exporting country flag	true
5	Importing Country	Boolean	n1	M	Importing country flag	false

Error Codes:

- COUNTRY_E001: "Invalid format" (HTTP 400)
- COUNTRY_E002: "Not approved for trade" (HTTP 422)
- COUNTRY_E003: "Sanctioned country" (HTTP 403)

Sample JSON Request:

```
curl 'https://uat.cncangola.com/api/Countries?$sort=Country_Name&Country_Name=&ExportingCountry=true&active=1' \
-H 'Accept: application/json, text/plain, */*' \
-H 'Accept-Language: en-US,en;q=0.9' \
-H 'Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9
```

Sample JSON Response:

```
[
  {
    "Id": 4,
    "Country_Code": "AF",
    "Country_Name": "Afghanistan",
    "Import_Fixed_Visa_Cost": false,
    "CountryZoneId": 3,
    "ContinentId": 3,
    "VisaCurrencyId": 2,
    "Export_Fixed_Visa_Cost": false,
```

```

"Enable_Discharge": false,
"MarkedForUpdate": false,
"IsCtnCountry": false,
"ExportingCountry": true,
"ImportingCountry": false,
"CountryZone": null,
"Continent": null,
"VisaCurrency": null,
"CreatedOn": null,
"CreatedById": null,
"CreatedByLogin": null,
"ModifiedOn": null,
"ModifiedById": null,
"ModifiedByLogin": null
}

```

5.1.4 Carriers API

Business Purpose: Manages shipping carrier information essential for maritime transport documentation and tracking. Critical for bill of lading validation, carrier liability determination, vessel scheduling coordination, and ensuring proper documentation of transportation responsibility throughout the supply chain.

Endpoint: GET /api/Carriers

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	\$sort	String	an..100	O	OData sort parameter	Name
2	Name	String	an..200	O	Filter by carrier name	A.C. ORSSLEFF'S
3	active	Boolean	n1	O	Filter active records only	1

UI Validation Rules:

- CARRIER_UI_001: Name filter optional, 1-200 characters
- CARRIER_UI_002: Display carrier name with city/country for clarity
- CARRIER_UI_003: Auto-complete dropdown with carrier search

Business Validation Rules:

- CARRIER_BV_001: Only active carriers allowed for new shipments
- CARRIER_BV_002: Carrier must be authorized for the route

- CARRIER_BV_003: Carrier contact information required for tracking

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	Id	Integer	n..10	M	Unique carrier identifier	789
2	Name	String	an..200	M	Carrier company name	A.C. ORSSLEFF'S EFTF A/S
3	LicenseNumber	String	an..50	O	Carrier license number	LICENSE123
4	Address	String	an..500	O	Carrier address	Kongevejen 40
5	City	String	an..100	O	Carrier city	2840 Holte
6	CountryId	Integer	n..10	O	Country identifier	56
7	Email	String	an..200	O	Contact email	chartering@acoe.dk
8	Telephone	String	an..50	O	Contact telephone	+45 45 46 00 66
9	TrackingURL	String	an..500	O	Tracking website URL	https://track.carrier.com

Error Codes:

- **CARRIER_E001:** "Invalid parameters" (HTTP 400)
- **CARRIER_E002:** "No data found" (HTTP 404)
- **CARRIER_E003:** "Access denied" (HTTP 403)

Sample JSON Request:

```
GET /api/Carriers?$sort=Name&Name=&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1343,
    "LicenseNumber": null,
```

```

    "Address": "Care of ZEABORN Ship Management GmbH & Cie KG , Ludwig-Erhard
-Strasse 22, 20459 Hamburg, Germany",
    "City": "Hamburg",
    "CountryId": 54,
    "Email": null,
    "Name": "\"E.R. BRISTOL\" Schiffsbeteiligungsgesellschaft",
    "Telephone": null,
    "TrackingURL": null,
    "CreatedOn": "2020-07-17T08:57:52.973",
    "CreatedById": 1,
    "ModifiedOn": "2020-07-17T08:57:52.973",
    "ModifiedById": 1
  },
  {
    "Id": 789,
    "LicenseNumber": null,
    "Address": "Kongevej 40",
    "City": "2840 Holte",
    "CountryId": 56,
    "Email": "chartering@acoe.dk",
    "Name": "A.C. ORSSLEFF'S EFTF A/S",
    "Telephone": "+45 45 46 00 66",
    "TrackingURL": "",
    "CreatedOn": "2008-05-13T08:36:14",
    "CreatedById": 1,
    "ModifiedOn": "2018-08-29T09:24:35.927",
    "ModifiedById": 1
  }
]

```

5.1.5 Currencies API

Endpoint: GET /api/Currencies

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	\$sort	String	an..100	O	OData sort parameter	Code
2	Code	String	a3	O	Filter by currency code	USD
3	active	Boolean	n1	O	Filter active records only	1

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	Id	Integer	n..10	M	Unique currency identifier	2
2	Code	String	a3	M	ISO currency code	USD
3	Name	String	an..100	O	Currency name	US Dollar
4	Symbol	String	an..5	O	Currency symbol	\$

Sample JSON Request:

```
GET /api/Currencies?$sort=Code&Code=USD&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "Code": "EUR",
    "MarkedForUpdate": false,
    "Countries": [],
    "Exchange_rates": [],
    "CreatedOn": null,
    "CreatedById": null,
    "ModifiedOn": null,
    "ModifiedById": null
  },
  {
    "Id": 2,
    "Code": "USD",
    "MarkedForUpdate": false,
    "Countries": [],
    "Exchange_rates": [],
    "CreatedOn": null,
    "CreatedById": null,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]
```

5.1.6 Banks API

Business Purpose: Manages banking institution data for payment processing and invoice generation in CNCA certificate transactions. Essential for financial settlements, payment validations, and generating proper invoicing documentation with correct banking details.

Endpoint: GET /api/Banks

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	\$sort	String	an..100	O	OData sort parameter	Name
2	Name	String	an..200	O	Filter by bank name	Standard Bank
3	active	Boolean	n1	O	Filter active records only	1

UI Validation Rules:

- BANK_UI_001: Bank name filter optional, 1-200 characters
- BANK_UI_002: Display active banks only in dropdown selection
- BANK_UI_003: Validate bank selection for payment processing

Business Validation Rules:

- BANK_BV_001: Only active banks allowed for new payments
- BANK_BV_002: Bank must support international transfers for foreign traders
- BANK_BV_003: Bank details required for invoice generation

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	Id	Integer	n..10	M	Unique bank identifier	15
2	Name	String	an..200	M	Bank institution name	Standard Bank Angola
3	Code	String	an..20	O	Bank code	SBA
4	SwiftCode	String	an..11	O	International SWIFT code	SBICAOLU
5	Active	Boolean	n1	M	Active status indicator	true

Error Codes:

- BANK_E001: "Invalid parameters" (HTTP 400)
- BANK_E002: "No data found" (HTTP 404)
- BANK_E003: "Access denied" (HTTP 403)

Sample JSON Request:

```
GET /api/Banks?$sort=Name&Name=&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "Name": "Banco Angolano de Investimentos",
    "Code": "BAI",
    "Address": "Rua Major Kanhangulo, nº 34",
    "City": "Luanda",
    "CountryID": 10,
    "Country": {
      "Id": 10,
      "Country_Code": "AO",
      "Country_Name": "Angola",
      "ExportingCountry": true,
      "ImportingCountry": true
    },
    "Website": "http://www.bancobai.ao",
    "Telephone": "222 693 800 / 222 693 899",
    "NIFNumber": null,
    "CreatedOn": null,
    "ModifiedOn": "2023-01-24T11:54:44.7181885",
    "ModifiedById": 1,
    "ModifiedByLogin": "TCNT"
  },
  {
    "Id": 2,
    "Name": "Banco Angolano de Negocios e Comercio, S.A.",
    "Code": "BANC",
    "Address": "Travessa da Sorte, nº 12. Maianga",
    "City": "Luanda",
    "CountryID": 10,
    "Country": {
      "Id": 10,
      "Country_Code": "AO",
      "Country_Name": "Angola",
      "ExportingCountry": true,
      "ImportingCountry": true
    },
    "Website": null,
    "Telephone": null,
    "NIFNumber": null,
    "CreatedOn": null,
    "ModifiedOn": null,
  }
]
```

```
"ModifiedById": null
}
```

- BANK_E002: “Bank not authorized for transactions” (HTTP 422)
- BANK_E003: “Inactive bank selected” (HTTP 422)

5.1.7 Units API

Business Purpose: Manages measurement unit definitions essential for accurate quantity and weight calculations in cargo documentation. Critical for ensuring consistent unit conversions, compliance with international standards, and accurate cost calculations based on cargo measurements.

Endpoint: GET /api/Units

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	\$sort	String	an..100	O	OData sort parameter	Unit_Desc
2	Unit_Description	String	an..50	O	Filter by unit description	KG
3	UnitType	String	an..20	O	Filter by unit category	Weight
4	active	Boolean	n1	O	Filter active records only	1

UI Validation Rules:

- - UNIT_UI_001: Unit description required, 1-50 characters
- - UNIT_UI_002: Unit type selection from predefined categories
- - UNIT_UI_003: Display with conversion factors for clarity

Business Validation Rules:

- - UNIT_BV_001: Unit must be internationally recognized standard
- - UNIT_BV_002: Conversion factors must be mathematically accurate
- - UNIT_BV_003: Weight units required for cargo calculations

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	Id	Integer	n..10	M	Unique unit identifier	1

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
2	Unit_Desc	String	an..50	M	Unit description	Kilogram
3	UnitSymbol	String	an..10	M	Standard unit symbol	KG
4	UnitType	String	an..20	M	Unit category	Weight
5	ConversionFactor	Decimal	n..10,6	O	Conversion to base unit	1.000
6	Active	Boolean	n1	M	Active status indicator	true

Error Codes:

- UNIT_E001: "Invalid unit parameters" (HTTP 400)
- UNIT_E002: "Unit not found" (HTTP 404)
- UNIT_E003: "Unit conversion error" (HTTP 422)

Sample JSON Request:

```
GET /api/Units?$sort=Unit_Desc&UnitType=Weight&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "Unit_Desc": "Kilogram",
    "UnitSymbol": "KG",
    "UnitType": "Weight",
    "ConversionFactor": 1.000,
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  },
  {
    "Id": 2,
    "Unit_Desc": "Ton",
    "UnitSymbol": "TON",
    "UnitType": "Weight",
    "ConversionFactor": 1000.000,
    "Active": true,
```

```

    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]

```

5.1.8 Container Types API

Business Purpose: Manages container type classifications essential for proper container booking, handling procedures, and shipping cost calculations. Critical for determining container capacity, weight limits, handling requirements, and ensuring accurate freight calculations based on container specifications.

Endpoint: GET /api/ContainerTypes

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	\$sort	String	an..100	O	OData sort parameter	Container_Type
2	Container_Type	String	an..50	O	Filter by container type	20GP
3	active	Boolean	n1	O	Filter active records only	1

UI Validation Rules:

- CONTAINER_UI_001: Container type filter optional, 1-50 characters
- CONTAINER_UI_002: Display container type with dimensions for clarity
- CONTAINER_UI_003: Group standard vs special container types

Business Validation Rules:

- CONTAINER_BV_001: Container type must match cargo requirements
- CONTAINER_BV_002: Dangerous goods require specialized containers
- CONTAINER_BV_003: Weight limits vary by container type

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	Id	Integer	n..10	M	Unique container type identifier	1

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
2	Container_Type	String	an..50	M	Container type code	20GP
3	Description	String	an..200	M	Container description	20-foot General Purpose
4	Length	Decimal	n..5,2	O	Container length in feet	20.00
5	Width	Decimal	n..5,2	O	Container width in feet	8.00
6	Height	Decimal	n..5,2	O	Container height in feet	8.50
7	MaxWeight	Decimal	n..8,2	O	Maximum weight capacity in tons	28.20
8	Active	Boolean	n1	M	Active status indicator	true

Error Codes:

- CONTAINER_E001: "Invalid parameters" (HTTP 400)
- CONTAINER_E002: "Container type not found" (HTTP 404)
- CONTAINER_E003: "Container type not applicable for cargo" (HTTP 422)

Sample JSON Request:

```
GET /api/ContainerTypes?$sort=Container_Type&Container_Type=20GP&active=1
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "Container_Type": "20GP",
    "Description": "20-foot General Purpose",
    "Length": 20.00,
    "Width": 8.00,
    "Height": 8.50,
    "MaxWeight": 28.20,
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
  }
]
```

```

    "ModifiedById": null
  },
  {
    "Id": 2,
    "Container_Type": "40GP",
    "Description": "40-foot General Purpose",
    "Length": 40.00,
    "Width": 8.00,
    "Height": 8.50,
    "MaxWeight": 30.48,
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]

```

5.1.9 Transport Types API

Business Purpose: Manages transportation mode classifications essential for logistics planning, route optimization, and freight calculations. Critical for determining shipping costs, transit times, regulatory requirements, and ensuring proper transport documentation based on cargo and route characteristics.

Endpoint: GET /api/TransportTypes

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	\$sort	String	an..100	O	OData sort parameter	Type_Name
2	Type_Name	String	an..50	O	Filter by transport type	Sea
3	active	Boolean	n1	O	Filter active records only	1

UI Validation Rules:

- TRANSPORT_UI_001: Transport type filter optional, 1-50 characters
- TRANSPORT_UI_002: Display with appropriate transport icons
- TRANSPORT_UI_003: Group by transport mode for clarity

Business Validation Rules:

- TRANSPORT_BV_001: Transport type must match cargo requirements

- TRANSPORT_BV_002: Dangerous goods require specific transport modes
- TRANSPORT_BV_003: Some goods restricted for certain transport types

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	Id	Integer	n..10	M	Unique transport type identifier	1
2	Type_Name	String	an..50	M	Transport type name	Sea
3	Description	String	an..200	O	Transport type description	Maritime sea transport
4	Code	String	an..10	O	Transport type code	SEA
5	Active	Boolean	n1	M	Active status indicator	true

Error Codes:

- TRANSPORT_E001: "Invalid parameters" (HTTP 400)
- TRANSPORT_E002: "Transport type not found" (HTTP 404)
- TRANSPORT_E003: "Transport type not applicable for cargo" (HTTP 422)

Sample JSON Request:

```
GET /api/TransportTypes?$sort=Type_Name&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "Type_Name": "Air",
    "Description": "Air cargo transport",
    "Code": "AIR",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  },
  {
    "Id": 2,
```

```

    "Type_Name": "Sea",
    "Description": "Maritime sea transport",
    "Code": "SEA",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedBy": 1,
    "ModifiedOn": null,
    "ModifiedBy": null
  }
]

```

5.1.10 Locations/Ports API

Business Purpose: Manages geographical location and port information essential for origin/destination tracking, routing decisions, and customs compliance. Critical for calculating shipping routes, determining applicable regulations, port charges, and ensuring accurate geographic references in certificates and shipping documents.

Endpoint: GET /api/Locations

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	\$sort	String	an..100	O	OData sort parameter	PortName
2	CountryId	Integer	n..10	M	Country identifier filter	4
3	PortName	String	an..200	O	Filter by port name	Dubai
4	active	Boolean	n1	O	Filter active records only	1

UI Validation Rules:

- LOCATION_UI_001: Country ID required for location filtering
- LOCATION_UI_002: Port name search minimum 2 characters
- LOCATION_UI_003: Display with country name for clarity

Business Validation Rules:

- LOCATION_BV_001: Location must be active port for shipping
- LOCATION_BV_002: Port must support container operations
- LOCATION_BV_003: Location must have valid customs facilities

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	Id	Integer	n..10	M	Unique location identifier	1
2	PortName	String	an..200	M	Port/location name	Port of Dubai
3	PortCode	String	an..10	O	Standard port code	DXB
4	CountryId	Integer	n..10	M	Associated country	4
5	CountryName	String	an..100	O	Country name	United Arab Emirates
6	Active	Boolean	n1	M	Active status indicator	true

Error Codes:

- LOCATION_E001: "Invalid country ID" (HTTP 400)
- LOCATION_E002: "Port not found" (HTTP 404)
- LOCATION_E003: "Port not operational for cargo" (HTTP 422)

Sample JSON Request:

```
GET /api/Locations?${sort=PortName&CountryId=4&active=1}
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "PortName": "Port of Dubai",
    "PortCode": "DXB",
    "CountryId": 4,
    "CountryName": "United Arab Emirates",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  },
  {
    "Id": 2,
    "PortName": "Abu Dhabi Port",
    "PortCode": "AUH",
    "CountryId": 4,
```

```

    "CountryName": "United Arab Emirates",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]

```

5.1.11 Goods Classifications API

Business Purpose: Manages standardized goods classification codes essential for customs clearance, duty calculations, and regulatory compliance. Critical for determining applicable tariffs, import restrictions, statistical reporting, and ensuring proper cargo categorization according to international trade standards.

Endpoint: GET /api/GoodsClassifications

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	\$sort	String	an..100	O	OData sort parameter	Goods_Classification_Desc
2	Goods_Classification_Desc	String	an..200	O	Filter by classification	Electronics
3	ClassificationCode	String	an..20	O	Filter by classification code	HS8517
4	active	Boolean	n1	O	Filter active records only	1

UI Validation Rules:

- - GOODS_UI_001: Classification filter minimum 3 characters
- - GOODS_UI_002: Display with HS code for clarity
- - GOODS_UI_003: Group by classification category

Business Validation Rules:

- - GOODS_BV_001: Classification must be valid HS code
- - GOODS_BV_002: Some goods require special permits
- - GOODS_BV_003: Prohibited goods cannot be classified

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	Id	Integer	n..10	M	Unique classification identifier	1
2	Goods_Classification_Desc	String	an..200	M	Classification description	Electronic equipment
3	ClassificationCode	String	an..20	M	HS classification code	HS8517
4	Category	String	an..100	O	Classification category	Electronics
5	DutyRate	Decimal	n..5,2	O	Applicable duty rate percentage	5.00
6	Active	Boolean	n1	M	Active status indicator	true

Error Codes:

- GOODS_E001: "Invalid classification parameters" (HTTP 400)
- GOODS_E002: "Classification not found" (HTTP 404)
- GOODS_E003: "Classification not applicable" (HTTP 422)

Sample JSON Request:

```
GET /api/GoodsClassifications?$sort=Goods_Classification_Desc&active=1
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "Goods_Classification_Desc": "Electronic equipment",
    "ClassificationCode": "HS8517",
    "Category": "Electronics",
    "DutyRate": 5.00,
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]
```

```

    },
    {
      "Id": 2,
      "Goods_Classification_Desc": "Textile products",
      "ClassificationCode": "HS6203",
      "Category": "Textiles",
      "DutyRate": 12.00,
      "Active": true,
      "CreatedOn": "2023-01-15T10:30:00Z",
      "CreatedById": 1,
      "ModifiedOn": null,
      "ModifiedById": null
    }
  ]

```

5.1.12 IMO Codes API

Business Purpose: Manages International Maritime Organization dangerous goods classification codes essential for hazardous cargo handling, safety compliance, and regulatory adherence. Critical for determining special handling requirements, storage restrictions, transport documentation, and ensuring maritime safety standards compliance.

Endpoint: GET /api/IMOs

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	\$sort	String	an..100	O	OData sort parameter	IMO_Desc
2	IMO_Desc	String	an..200	O	Filter by IMO description	Flammabl e Liquids
3	IMOCClass	String	an..10	O	Filter by IMO class	3
4	active	Boolean	n1	O	Filter active records only	1

UI Validation Rules:

- IMO_UI_001: IMO description filter minimum 3 characters
- IMO_UI_002: Display with hazard class for safety
- IMO_UI_003: Highlight dangerous goods with warning indicators

Business Validation Rules:

- IMO_BV_001: IMO class must be valid IMDG code

- - IMO_BV_002: Dangerous goods require special documentation
- - IMO_BV_003: Some IMO classes prohibited for certain routes

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	Id	Integer	n..10	M	Unique IMO identifier	1
2	IMO_Desc	String	an..200	M	IMO description	Flammable Liquids
3	IMOCClass	String	an..10	M	IMDG hazard class	3
4	UNNumber	String	an..10	O	UN identification number	UN1203
5	PackingGroup	String	an..5	O	Packing group classification	II
6	Active	Boolean	n1	M	Active status indicator	true

Error Codes:

- IMO_E001: "Invalid IMO parameters" (HTTP 400)
- IMO_E002: "IMO code not found" (HTTP 404)
- IMO_E003: "Dangerous goods not permitted" (HTTP 422)

Sample JSON Request:

```
GET /api/IMOs?$sort=IMO_Desc&IMOCClass=3&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "IMO_Desc": "Flammable Liquids",
    "IMOCClass": "3",
    "UNNumber": "UN1203",
    "PackingGroup": "II",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  },
]
```

```
{
  "Id": 2,
  "IMO_Desc": "Corrosive Substances",
  "IMOCClass": "8",
  "UNNumber": "UN1760",
  "PackingGroup": "II",
  "Active": true,
  "CreatedOn": "2023-01-15T10:30:00Z",
  "CreatedById": 1,
  "ModifiedOn": null,
  "ModifiedById": null
}
```

5.1.13 Vessels API

Business Purpose: Manages vessel information essential for maritime transport operations, vessel tracking, and shipping documentation. Critical for identifying vessels in certificates, tracking cargo movements, ensuring vessel compliance, and providing accurate shipping details for customs and logistics operations.

Endpoint: GET /api/Vessels

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	\$sort	String	an..100	O	OData sort parameter	Name
2	Name	String	an..256	O	Filter by vessel name	CPO HAMBURG
3	Code	String	an..20	O	Filter by vessel code	229638000
4	active	Integer	n1	O	Active status filter	1

UI Validation Rules:

- VESSEL_UI_001: Vessel name filter minimum 3 characters
- VESSEL_UI_002: Display with IMO number for identification
- VESSEL_UI_003: Show flag state for vessel information

Business Validation Rules:

- VESSEL_BV_001: Vessel must have valid IMO number
- VESSEL_BV_002: Vessel must be registered and active
- VESSEL_BV_003: Vessel must be suitable for cargo type

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	Id	Integer	n..10	M	Unique vessel identifier	5132
2	Name	String	an..256	M	Vessel name	CPO HAMBURG 1234
3	Code	String	an..20	M	Vessel identification code	229638000
4	IDNumber	String	an..20	M	IMO number	9450375
5	FlagId	Integer	n..10	O	Flag state identifier	148
6	Flag	String	an..100	O	Flag state name	Germany
7	Active	Boolean	n1	M	Active status	true

Error Codes:

- VESSEL_E001: "Invalid vessel parameters" (HTTP 400)
- VESSEL_E002: "Vessel not found" (HTTP 404)
- VESSEL_E003: "Vessel not available for route" (HTTP 422)

Sample JSON Request:

```
GET /api/Vessels?$sort=Name&Name=CP0&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 5132,
    "Name": "CPO HAMBURG1234",
    "Code": "229638000",
    "IDNumber": "9450375",
    "FlagId": 148,
    "Flag": "Germany",
    "Active": true,
    "CreatedOn": null,
    "CreatedById": null,
    "CreatedByLogin": null,
    "ModifiedOn": "2023-01-25T13:18:15.1194343",
    "ModifiedById": 13275,
    "ModifiedByLogin": "TESTEXCONSUL"
```

```

    },
    {
      "Id": 5163,
      "Name": "NILEDUTCH LION",
      "Code": "636013690",
      "IDNumber": "9337456",
      "FlagId": 128,
      "Flag": "Liberia",
      "Active": true,
      "CreatedOn": null,
      "CreatedById": null,
      "CreatedByLogin": null,
      "ModifiedOn": null,
      "ModifiedById": null,
      "ModifiedByLogin": null
    }
  ]

```

5.1.14 CTN Cities API

Business Purpose: Manages city information essential for origin and destination tracking in certificate processing. Critical for accurate geographical references, determining applicable regulations, calculating logistics routes, and ensuring precise location data for customs compliance and certificate validation.

Endpoint: GET /api/CTNCities

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	\$sort	String	an..100	O	OData sort parameter	Name
2	CountryId	Integer	n..10	M	Country identifier filter	10
3	Name	String	an..200	O	Filter by city name	Luanda
4	active	Boolean	n1	O	Filter active records only	1

UI Validation Rules:

- CITY_UI_001: Country ID required for city filtering
- CITY_UI_002: City name search minimum 2 characters
- CITY_UI_003: Display with country name for clarity

Business Validation Rules:

- - CITY_BV_001: City must exist within selected country
- - CITY_BV_002: City must be active for trade operations
- - CITY_BV_003: City must support logistics operations

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	Id	Integer	n..10	M	Unique city identifier	262
2	Name	String	an..200	M	City name	Benguela
3	CountryId	Integer	n..10	M	Associated country identifier	10
4	Country	String	an..100	O	Country name	Angola
5	Active	Boolean	n1	M	Active status indicator	true

Error Codes:

- CITY_E001: "Invalid country ID" (HTTP 400)
- CITY_E002: "City not found" (HTTP 404)
- CITY_E003: "City not operational for trade" (HTTP 422)

Sample JSON Request:

```
GET /api/CTNCities?$sort=Name&CountryId=10&active=1
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 262,
    "Active": true,
    "CountryId": 10,
    "Name": "Benguela",
    "Country": "Angola",
    "CreatedOn": null,
    "CreatedById": null,
    "CreatedByLogin": null,
    "ModifiedOn": null,
    "ModifiedById": null,
    "ModifiedByLogin": null
  },
  {
```

```

    "Id": 277,
    "Active": true,
    "CountryId": 10,
    "Name": "Caála",
    "Country": "Angola",
    "CreatedOn": null,
    "CreatedById": null,
    "CreatedByLogin": null,
    "ModifiedOn": null,
    "ModifiedById": null,
    "ModifiedByLogin": null
  },
  {
    "Id": 255,
    "Active": true,
    "CountryId": 10,
    "Name": "Cabinda",
    "Country": "Angola",
    "CreatedOn": null,
    "CreatedById": null,
    "CreatedByLogin": null,
    "ModifiedOn": null,
    "ModifiedById": null,
    "ModifiedByLogin": null
  }
]

```

5.1.15 CTN Ports API

Business Purpose: Manages Angolan port information essential for CTN certificate processing and customs clearance. Critical for determining applicable port authorities, customs offices, handling procedures, and ensuring accurate port references in certificates and logistics documentation.

Endpoint: GET /api/CTNPorts

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	\$sort	String	an..100	O	OData sort parameter	Name
2	Name	String	an..100	O	Filter by port name	Luanda
3	PortCode	String	an..10	O	Filter by port code	LAD
4	active	Boolean	n1	O	Filter active records only	1

UI Validation Rules:

- - CTNPORT_UI_001: Port name search minimum 2 characters
- - CTNPORT_UI_002: Display with customs office information
- - CTNPORT_UI_003: Show operational status clearly

Business Validation Rules:

- - CTNPORT_BV_001: Port must be operational for CTN processing
- - CTNPORT_BV_002: Customs office must be active
- - CTNPORT_BV_003: Port must support container operations

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	Id	Integer	n..10	M	Unique port identifier	1
2	Name	String	an..100	M	Port name	Port of Luanda
3	PortCode	String	an..10	M	Standard port code	LAD
4	CustomsOffice	String	an..100	O	Associated customs office	Alfândega de Luanda
5	Active	Boolean	n1	M	Active status indicator	true

Error Codes:

- CTNPORT_E001: "Invalid port parameters" (HTTP 400)
- CTNPORT_E002: "Port not found" (HTTP 404)
- CTNPORT_E003: "Port not operational" (HTTP 422)

Sample JSON Request:

```
GET /api/CTNPorts?$sort=Name&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
    "Name": "Port of Luanda",
```

```

    "PortCode": "LAD",
    "CustomsOffice": "Alfândega de Luanda",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  },
  {
    "Id": 2,
    "Name": "Port of Lobito",
    "PortCode": "LBT",
    "CustomsOffice": "Alfândega de Lobito",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]

```

Business Purpose: Manages CTN certificate attachments and supporting documents essential for compliance verification and audit trails. Critical for storing required documentation, enabling file downloads, maintaining document integrity, and ensuring complete certificate packages for customs and regulatory review.

Endpoint: GET /api/ctnAttachments

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	\$expand	String	an..500	O	OData expand parameter	CreatedBy, attachment, attachmentName
2	ctn	String	an..100	M	CTN identifier	503808
3	ctnid	Integer	n..3	M	CTN identifier (alternative)	503808

UI Validation Rules:

- CTNATT_UI_001: CTN ID required for attachment lookup
- CTNATT_UI_002: Display file types with appropriate icons
- CTNATT_UI_003: Show file size and upload date for management

Business Validation Rules:

- - CTNATT_BV_001: CTN must exist and be accessible
- - CTNATT_BV_002: File size must not exceed system limits
- - CTNATT_BV_003: File types must be approved formats

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Format/Derivation Logic	Data Example
1	Id	Integer	n..10	M	Attachment record ID	12345
2	CTNId	String	an..50	M	Related CTN ID	503808
3	FileName	String	an..50	M	Original file name	bill_of_lading.pdf
4	FileSize	String	an..50	M	File size in bytes	2048576
5	AttachmentType	String	an..50	M	Type of attachment	1
6	AttachmentType Name	Integer	n..10	O	Attachment type description	Bill of Lading
7	UploadDate	Integer	n..10	M	File upload timestamp	2023-12-15T10:30:00Z
8	CreatedBy	Data Type	Format	M	User who uploaded file	13345

Error Codes:

- CTNATT_E001: “Invalid CTN ID” (HTTP 400)
- CTNATT_E002: “No attachments found” (HTTP 404)
- CTNATT_E003: “File access denied” (HTTP 403)

Sample JSON Request:

```
GET /api/ctnAttachments?$expand=CreatedBy,attachment,attachmentName&ctnid=503808
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 12345,
    "CTNId": 503808,
    "FileName": "bill_of_lading.pdf",
    "FileSize": 2048576,
    "AttachmentTypeId": 1,
    "AttachmentTypeName": "Bill of Lading",
    "UploadDate": "2023-12-15T10:30:00Z",
    "CreatedById": 13345,
    "CreatedBy": {
      "Id": 13345,
      "UserName": "CNCEXPORER",
      "Email": "exporter@company.com"
    }
  }
]
```

5.2 Certificate Management APIs

Certificate management APIs handle the core CTN certificate lifecycle operations, enabling the creation, retrieval, and management of CNCA certificates throughout their entire lifecycle from draft to completion.

5.2.2 CTN Creation API (Certificate Submission)

Note: This unified API consolidates what was previously multiple separate operations (creation, addresses, goods, containers, tracking, attachments) into one comprehensive submission. JUL will handle all data processing internally and save as draft, eliminating the need for multiple API calls from SINTECE.

Business Purpose: Creates new CTN certificate records with complete information for CNCA certificate applications in a single comprehensive request. This unified API combines what was previously multiple separate operations (creation, addresses, goods, containers, tracking, attachments) into one streamlined submission. Critical for submitting complete certificate requests, establishing certificate records with all required data, validating comprehensive submission data, and enabling the certificate approval workflow efficiently.

Endpoint: POST /api/ctns

Request Elements (Core CTN Information):

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	Id	Integer	n..10	O	CTN identifier (null for new records)	null
2	CTN_Reference_Number	String	an..50	O	System generated reference (auto-assigned)	“”
3	StatusId	Integer	n..10	M	Status identifier (1=Created/Draft)	1
4	Groupage	Boolean	n1	O	Groupage indicator (true/false)	false
5	ParentCTNId	Integer	n..10	O	Parent CTN identifier (null for standalone)	null
6	CargoTypeId	Integer	n..10	M	Cargo type identifier (foreign key, must be valid)	1
7	ETD	DateTime	DATE	O	Estimated time departure (ISO 8601, nullable)	null
8	ETA	DateTime	DATE	O	Estimated time arrival (ISO 8601, nullable)	null
9	BL_number	String	an..50	M	Bill of lading number (unique, alphanumeric)	vc568009iujh
10	IncotermId	Integer	n..10	M	Incoterm identifier (foreign key, must be valid)	2
11	OriginCountryId	Integer	n..10	M	Origin country identifier (foreign key, must be valid)	10

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
12	FinalDestinationCountryId	Integer	n..10	M	Destination country identifier (foreign key)	4
13	FreightPaymentTypeId	Integer	n..10	M	Freight payment type identifier (foreign key)	1
14	Total_number_containers	Integer	n..5	O	Total containers count (positive integer)	0
15	Total_number_vehicles	Integer	n..5	O	Total vehicles count (positive integer)	0
16	Total_Ocean_Freight	Decimal	n..16,3	O	Total ocean freight value (max 3 decimals)	0
17	Total_Value_Of_Goods	Decimal	n..16,3	O	Total goods value in USD (max 3 decimals)	0
18	Total_Charges	Decimal	n..16,3	O	Total charges amount (max 3 decimals)	0
19	General_Total	Decimal	n..16,3	O	General total amount (max 3 decimals)	0
20	View_CurrencyId	Integer	n..10	M	Currency identifier (foreign key, must be valid)	1
21	Exchange_Rate	Decimal	n..12,5	O	Exchange rate (max 5 decimals)	1
22	VoyageNo	String	an..50	O	Voyage number (alphanumeric)	null
23	IsExport	Boolean	n1	M	Export indicator (true for export, false for import)	true
24	IsImport	Boolean	n1	O	Import indicator (true/false)	false

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
25	Origin_City Id	Integer	n..10	M	Origin city identifier (foreign key, must be valid)	262
26	Final_Destination_City Id	Integer	n..10	O	Final destination city (foreign key, nullable)	null
27	CarrierId	Integer	n..10	M	Carrier identifier (foreign key, must be valid)	789
28	UniqueTradeNumber	String	an..50	M	Unique trade number/DUP (unique, numeric)	56789098765
29	BankId	Integer	n..10	O	Bank identifier (foreign key, nullable)	2
30	ConsigneeId	Integer	n..10	M	Consignee identifier (foreign key, must be valid)	43214
31	DCNumber	String	an..50	O	Document control number (alphanumeric)	7777777
32	CreatedOn	DateTime	DATE	O	Creation timestamp (ISO 8601, system generated)	2025-11-12T18:49:30.88Z
33	CreatedById	Integer	n..10	O	Creator identifier (system generated)	1

Request Elements (Address Information - Array):

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
34	CTN_Addresses	Array	-	O	Array of address objects (shipper, consignee, forwarder, notify)	[...]

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
34.1	Id	Integer	n..10	O	Address identifier (null for new records)	null
34.2	QryAddressTypeId	Integer	n..10	M	Address type (1=Shipper, 2=Consignee, 3=Forwarder, 4=Notify)	2
34.3	Name	String	an..200	M	Company/Person name (max 200 chars)	M.A.C.S AS.TEC.N. E IND
34.4	Address	String	an..500	M	Physical address (max 500 chars)	maculusso rua comandate
34.5	City	String	an..100	M	City name (max 100 chars)	kwenhan
34.6	CountryId	Integer	n..10	M	Country identifier (foreign key, must be valid)	10
34.7	Email	String	an..100	O	Email address (valid email format)	contact@co mpany.com
34.8	Telephone	String	an..50	M	Phone number (max 50 chars)	923798669
34.9	NIFNumber	String	an..50	M	Tax identification number (9-14 digits)	0000000458 608

Request Elements (Goods Information - Array):

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
35	CTN_Goods	Array	-	O	Array of goods/cargo objects	[...]
35.1	Id	Integer	n..10	O	Goods identifier (null for new records)	null

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
35.2	GoodsClassificationId	Integer	n..10	M	Goods classification identifier (foreign key)	16882
35.3	IMOId	Integer	n..10	M	IMO hazardous code identifier (foreign key)	2
35.4	CargoTypeId	Integer	n..10	M	Cargo type identifier (foreign key)	1
35.5	DescriptionGoods	String	an..500	M	Goods description (max 500 chars)	test
35.6	GrossWeightKG	Decimal	n..16,6	M	Gross weight in kg (max 6 decimals, positive)	10.0
35.7	VolumeCBM	Decimal	n..16,6	M	Volume in cubic meters (max 6 decimals, positive)	11.0
35.8	OceanFreight	Decimal	n..16,3	M	Ocean freight cost (max 3 decimals, positive)	22.0
35.9	ValueOfGoods	Decimal	n..16,3	M	Value of goods in USD (max 3 decimals, positive)	33.0
35.10	NumberOfPackages	Integer	n..8	M	Number of packages (positive integer, no decimals)	3

Request Elements (Container Information - Array):

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
36	CTN_Containers	Array	-	O	Array of container objects	[...]
36.1	Id	Integer	n..10	O	Container identifier (null for new records)	null
36.2	ContainerTypeld	Integer	n..10	M	Container type identifier (foreign key, must be valid)	31
36.3	Number_of_Containers	Integer	n..5	M	Number of containers (positive integer)	1
36.4	Container Numbers	String	an..100	M	Container number(s) (alphanumeric, max 100 chars)	MSCU1234567
36.5	SealNumbers	String	an..100	M	Seal number(s) (alphanumeric, max 100 chars)	23234
36.6	Groupage	Boolean	n1	O	Groupage indicator (true/false)	false
36.7	IsEmpty	Boolean	n1	O	Empty container indicator (true/false)	false
36.8	OwnedByShipper	Boolean	n1	O	Shipper owned indicator (true/false)	false

Request Elements (Transport/Tracking Information - Array):

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
37	CTN_Tracking	Array	-	O	Array of transport route objects	[...]
37.1	Id	Integer	n..10	O	Tracking identifier (null for new records)	null

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
37.2	SourceCountryId	Integer	n..10	M	Source country identifier (foreign key, must be valid)	10
37.3	SourcePortId	Integer	n..10	M	Source port identifier (foreign key, must be valid)	80
37.4	ETD	DateTime	DATE	M	Estimated time departure (ISO 8601, must be before ETA)	2025-11-12T00:00:00.000Z
37.5	TransportTypeId	Integer	n..10	M	Transport type identifier (foreign key, must be valid)	1
37.6	VoyageNumber	String	an..50	M	Voyage number (alphanumeric, max 50 chars)	546894
37.7	CarrierId	Integer	n..10	M	Carrier identifier (foreign key, must be valid)	789
37.8	VesselId	Integer	n..10	M	Vessel identifier (foreign key, must be valid)	5163
37.9	DestinationCountryId	Integer	n..10	M	Destination country identifier (foreign key)	4
37.10	DestinationPortId	Integer	n..10	M	Destination port identifier (foreign key, must be valid)	35
37.11	ETA	DateTime	DATE	M	Estimated time arrival (ISO 8601, must be after ETD)	2025-12-09T00:00:00.000Z
37.12	Sequence	Integer	n..5	M	Sequence number (positive)	1

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
					integer for route ordering)	

Request Elements (Attachment Information - Array):

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
38	CTN_Attachments	Array	-	O	Array of document attachment objects	[...]
38.1	Id	Integer	n..10	O	Attachment identifier (null for new records)	null
38.2	AttachmentGuid	String	an..50	M	File upload GUID from fileupload API (UUID format)	0da7461c-2155-4d20-a695-6b7463367327
38.3	NameId	Integer	n..10	M	Attachment name type identifier (foreign key)	4

UI Validation Rules:

- CTNCRE_UI_001: All mandatory fields must be completed for core CTN information
- CTNCRE_UI_002: BL number must be unique and valid format
- CTNCRE_UI_003: Trade number must follow required pattern
- CTNCRE_UI_004: At least one address must be provided (typically consignee)
- CTNCRE_UI_005: All address fields must be complete when address is provided
- CTNCRE_UI_006: At least one goods entry must be provided for the shipment
- CTNCRE_UI_007: Container information must match declared container count
- CTNCRE_UI_008: Transport route must have valid origin and destination
- CTNCRE_UI_009: Attachment GUIDs must be valid from fileupload API

Business Validation Rules:

- - CTNCRE_BV_001: BL number must not exist in system
- - CTNCRE_BV_002: All referenced entities (countries, ports, carriers, etc.) must be valid and active
- - CTNCRE_BV_003: User must have permission to create CTNs

- - CTNCRE_BV_004: NIFNumber validation must pass for Angola addresses
- - CTNCRE_BV_005: Total values must match sum of individual goods values
- - CTNCRE_BV_006: Container count must match actual container entries
- - CTNCRE_BV_007: Transport sequence must be logical (ETD before ETA)
- - CTNCRE_BV_008: Required attachments must be present based on trade type

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	Id	Integer	n..10	M	New CTN identifier (system generated)	503808
2	CTN_Reference_Number	String	an..50	M	System generated reference (format: AO-CNT-{ID}-{YEAR})	AO-CNT-503808-2025
3	StatusId	Integer	n..10	M	Initial status (1=Draft, system assigned)	1
4	BL_number	String	an..50	M	Confirmed BL number (as submitted)	vc568009iujh
5	UniqueTradeNumber	String	an..50	M	Confirmed trade number (as submitted)	56789098765
6	CreatedOn	DateTime	DATE	M	Creation timestamp (ISO 8601, system generated)	2025-11-14T10:30:00Z
7	CreatedById	Integer	n..10	M	Creator user identifier (system generated)	1
8	Addresses Created	Integer	n..5	M	Number of addresses created (count)	2
9	GoodsCreated	Integer	n..5	M	Number of goods entries created (count)	1

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
10	Containers Created	Integer	n..5	M	Number of container entries created (count)	1
11	Tracking Created	Integer	n..5	M	Number of tracking entries created (count)	1
12	Attachments Created	Integer	n..5	M	Number of attachments created (count)	2

Error Codes:

- CTNCRE_E001: "BL number already exists" (HTTP 409)
- CTNCRE_E002: "Invalid reference data" (HTTP 400)
- CTNCRE_E003: "Insufficient permissions" (HTTP 403)
- CTNCRE_E004: "Invalid address information" (HTTP 422)
- CTNCRE_E005: "Invalid goods information" (HTTP 422)
- CTNCRE_E006: "Invalid container information" (HTTP 422)
- CTNCRE_E007: "Invalid tracking information" (HTTP 422)
- CTNCRE_E008: "Invalid attachment information" (HTTP 422)
- CTNCRE_E009: "NIF validation failed" (HTTP 422)

Sample HTTP Request:

```
POST /api/ctns
Content-Type: application/json
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
```

```
{
  "Id": null,
  "CTN_Reference_Number": "",
  "StatusId": 1,
  "Groupage": false,
  "ParentCTNId": null,
  "CargoTypeId": 1,
  "ETD": null,
  "ETA": null,
  "BL_number": "vc568009iujh",
  "IncotermId": 2,
  "OriginCountryId": 10,
  "FinalDestinationCountryId": 4,
  "FreightPaymentTypeId": 1,
```

```
"Total_number_containers": 1,
"Total_number_vehicles": 0,
"Total_Ocean_Freight": 22.0,
"Total_Value_Of_Goods": 33.0,
"Total_Charges": 55.0,
"General_Total": 55.0,
"View_CurrencyId": 1,
"Exchange_Rate": 1,
"Visum_Location": null,
"Visum_Reference_number": null,
"Visum_Cost": 1,
"Visum_Date": null,
"Visum_AgentId": null,
"Invoicer_to_agent": null,
"VoyageNo": "546894",
"RegularisationCTN": false,
"IsExport": true,
"IsImport": false,
"Origin_CityId": 262,
"Final_Destination_CityId": null,
"CarrierId": 789,
"UniqueTradeNumber": "56789098765",
"Commission": 0,
"CommissionCNC": 0,
"CTNExchangeRate": null,
"MarkedForUpdate": false,
"ReceivedByGumar": false,
"ErrorSendToGumar": false,
"BankId": 2,
"ConsigneeId": 43214,
"ReExport": false,
"DCNumber": "77777777",
"CreatedOn": "2025-11-14T10:30:00Z",
"CreatedById": 1,
"CTN_Addresses": [
  {
    "Id": null,
    "QryAddressTypeId": 1,
    "Name": "EXPORTER COMPANY LTD",
    "Address": "123 Export Street",
    "City": "Export City",
    "CountryId": 10,
    "Email": "exporter@company.com",
    "Telephone": "244123456789",
    "NIFNumber": "1234567890"
  },
  {
    "Id": null,
    "QryAddressTypeId": 2,
```

```

    "Name": "M.A.C.S AS.TEC.N. E IND",
    "Address": "maculusso rua comandate",
    "City": "kwenhan",
    "CountryId": 4,
    "Email": "contact@company.com",
    "Telephone": "923798669",
    "NIFNumber": "0000000458608"
  }
],
"CTN_Goods": [
  {
    "Id": null,
    "GoodsClassificationId": 16882,
    "IMOId": 2,
    "CargoTypeId": 1,
    "DescriptionGoods": "Electronic Components and Parts",
    "GrossWeightKG": 1000.0,
    "VolumeCBM": 11.0,
    "OceanFreight": 22.0,
    "ValueOfGoods": 33.0,
    "NumberOfPackages": 3
  }
],
"CTN_Containers": [
  {
    "Id": null,
    "ContainerTypeId": 31,
    "Number_of_Containers": 1,
    "ContainerNumbers": "MSCU1234567",
    "SealNumbers": "23234",
    "Groupage": false,
    "IsEmpty": false,
    "OwnedByShipper": false
  }
],
"CTN_Tracking": [
  {
    "Id": null,
    "SourceCountryId": 10,
    "SourcePortId": 80,
    "ETD": "2025-11-15T00:00:00.000Z",
    "TransportTypeId": 1,
    "VoyageNumber": "546894",
    "CarrierId": 789,
    "VesselId": 5163,
    "DestinationCountryId": 4,
    "DestinationPortId": 35,
    "ETA": "2025-12-09T00:00:00.000Z",
    "Sequence": 1
  }
]

```



```

    }
  ],
  "CTN_RORO": [],
  "CTN_Attachments": [
    {
      "Id": null,
      "AttachmentGuid": "0da7461c-2155-4d20-a695-6b7463367327",
      "NameId": 4
    },
    {
      "Id": null,
      "AttachmentGuid": "1eb8572d-3266-5e31-b7a6-7c8574468438",
      "NameId": 1
    }
  ]
}

```

Sample JSON Response:

```

{
  "Id": 503808,
  "CTN_Reference_Number": "AO-CNT-503808-2025",
  "StatusId": 1,
  "BL_number": "vc568009iujh",
  "UniqueTradeNumber": "56789098765",
  "CreatedOn": "2025-11-14T10:30:00Z",
  "CreatedById": 1,
  "AddressesCreated": 2,
  "GoodsCreated": 1,
  "ContainersCreated": 1,
  "TrackingCreated": 1,
  "AttachmentsCreated": 2,
  "Message": "CTN certificate created successfully with all associated data"
}

```

5.2.4 Request Visa API (Certificate Issuance Submission)

Business Purpose: Submits completed CTN certificates for approval and visa issuance essential for finalizing the certificate approval process. Critical for triggering certificate review, validating completeness requirements, initiating approval workflow, and moving certificates from draft to approval status.

Endpoint: POST /api/ctns/actions/requestvisa/{id}

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	id	Integer	n..10	M	CTN identifier from URL path (positive integer)	503808
2	Content-Length	Integer	n..10	M	Request body length (0 for this action endpoint)	0

Pre-submission Validation Requirements: - **Shipper Address:** Complete shipper address information required - **Forwarder Address:** Complete forwarder address information required

- **Attachments:** At least 1 attachment must be uploaded - **CTN Data:** All mandatory CTN fields must be completed

UI Validation Rules:

- REQVISA_UI_001: CTN must be in “Edited” status to request visa
- REQVISA_UI_002: Display validation errors clearly to user
- REQVISA_UI_003: Confirm all requirements before submission

Business Validation Rules:

- REQVISA_BV_001: CTN must have complete address information
- REQVISA_BV_002: Required attachments must be uploaded
- REQVISA_BV_003: CTN must pass all business validations

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	rowsAffected	Integer	n..10	M	Number of affected records (-1 for success)	-1
2	error	String	an..1000	O	Validation error messages (empty on success)	“”
3	newCtnId	Integer	n..10	M	CTN identifier (0 for existing CTN)	0

Error Codes:

- REQVISA_E001: “Missing or incomplete Shipper Address” (HTTP 400)
- REQVISA_E002: “Missing or incomplete Forwarder Address” (HTTP 400)

- REQVISA_E003: “At least 1 attachment must be uploaded” (HTTP 400)
- REQVISA_E004: “CTN not found or not accessible” (HTTP 404)

Sample HTTP Request:

```
POST /api/ctns/actions/requestvisa/503808
Content-Length: 0
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
{
  "rowsAffected": -1,
  "error": "",
  "newCtnId": 0
}
```

5.2.5.1 Document Upload API for Request Visa

Business Purpose: Enables SINTECE to upload documents specifically for the Request Visa process, providing document management capabilities separate from the general file upload API. Critical for visa application documentation, compliance verification, and regulatory submission requirements.

Endpoint: POST /api/ctns/{ctnId}/documents/upload

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	ctnId	Integer	n..10	M	CTN identifier from URL path, positive integer	503808
2	file	Binary	-	M	Document file uploaded via multipart/form-data	[binary file data]
3	document Type	String	an..50	M	Type of document being uploaded, must be from approved list	BILL_OF_LADING
4	document Name	String	an..256	M	Document name/title, max 256 characters	Bill of Lading Original

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
5	description	String	an..500	O	Document description, free text, max 500 characters	Original shipping document for visa application
6	isRequired	Boolean	n1	M	Boolean indicator whether document is mandatory (1=true, 0=false)	true

UI Validation Rules:

- DOCUP_UI_001: File size must not exceed 25MB for visa documents
- DOCUP_UI_002: Document type must be from approved list
- DOCUP_UI_003: Document name required, max 256 characters
- DOCUP_UI_004: File types allowed: PDF, JPG, PNG, DOC, DOCX, XLS, XLSX

Business Validation Rules:

- DOCUP_BV_001: CTN must exist and be accessible
- DOCUP_BV_002: Document type must be valid for visa application
- DOCUP_BV_003: User must have permission to upload documents
- DOCUP_BV_004: Duplicate documents should be detected

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	documentId	String (UUID)	an..50	M	Unique document identifier, UUID format, system generated	f47ac10b-58cc-4372-a567-0e02b2c3d479
2	documentName	String	an..256	M	Uploaded document name, max 256 characters	Bill of Lading Original

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
3	fileName	String	an..256	M	Original file name from upload, max 256 characters	BOL_ABC123.pdf
4	fileSize	Integer	n..10	M	File size in bytes, positive integer	1048576
5	mimeType	String	an..100	M	File MIME type (e.g., application/pdf, image/jpeg)	application/pdf
6	downloadUrl	String	an..500	M	Relative URL path to download document	/api/ctns/503808/documents/47ac10b-58cc-4372-a567-0e02b2c3d479/download
7	uploadDate	DateTime	DATE	M	Upload timestamp, ISO 8601 format, system generated	2025-11-14T15:30:00Z
8	documentType	String	an..50	M	Document type classification	BILL_OF_LADING
9	status	String	an..20	M	Document processing status (e.g., UPLOADED, VERIFIED)	UPLOADED
10	ctnId	Integer	n..10	M	Associated CTN identifier	503808

Error Codes:

- DOCUP_E001: "CTN not found" (HTTP 404)
- DOCUP_E002: "Invalid document type" (HTTP 400)
- DOCUP_E003: "File size exceeds limit" (HTTP 413)
- DOCUP_E004: "Invalid file type" (HTTP 415)
- DOCUP_E005: "Document upload failed" (HTTP 500)

Sample Multipart Request:

```

POST /api/ctns/503808/documents/upload
Content-Type: multipart/form-data; boundary=----WebKitFormBoundary7MA4YWxkTrZu0gW
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...

-----WebKitFormBoundary7MA4YWxkTrZu0gW
Content-Disposition: form-data; name="file"; filename="BOL_ABC123.pdf"
Content-Type: application/pdf

[Binary PDF file content]
-----WebKitFormBoundary7MA4YWxkTrZu0gW
Content-Disposition: form-data; name="documentType"

BILL_OF_LADING
-----WebKitFormBoundary7MA4YWxkTrZu0gW
Content-Disposition: form-data; name="documentName"

Bill of Lading Original
-----WebKitFormBoundary7MA4YWxkTrZu0gW
Content-Disposition: form-data; name="description"

Original shipping document for visa application
-----WebKitFormBoundary7MA4YWxkTrZu0gW
Content-Disposition: form-data; name="isRequired"

true
-----WebKitFormBoundary7MA4YWxkTrZu0gW--
  
```

Sample JSON Response:

```

{
  "documentId": "f47ac10b-58cc-4372-a567-0e02b2c3d479",
  "documentName": "Bill of Lading Original",
  "fileName": "BOL_ABC123.pdf",
  "fileSize": 1048576,
  "mimeType": "application/pdf",
  "downloadUrl": "/api/ctns/503808/documents/f47ac10b-58cc-4372-a567-0e02b2c3d479/download",
  "uploadDate": "2025-11-14T15:30:00Z",
  "documentType": "BILL_OF_LADING",
  "status": "UPLOADED",
  "ctnId": 503808
}
  
```

5.2.5.2 Document Delete API for Request Visa

Business Purpose: Allows SINTECE to delete uploaded documents from CTN certificates before visa submission. Critical for document management, correcting upload mistakes, replacing outdated documents, and maintaining clean documentation sets.

Endpoint: DELETE /api/ctns/{ctnId}/documents/{documentId}

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	ctnId	Integer	n..10	M	CTN identifier from URL path, positive integer	503808
2	documentId	String (UUID)	an..50	M	Document identifier from URL path, UUID format	f47ac10b-58cc-4372-a567-0e02b2c3d479
3	reason	String	an..200	O	Deletion reason in request body, free text, max 200 characters	Incorrect document uploaded

UI Validation Rules:

- - DOCDEL_UI_001: Confirm deletion with user before proceeding
- - DOCDEL_UI_002: Display document details before deletion
- - DOCDEL_UI_003: Reason for deletion optional but recommended

Business Validation Rules:

- - DOCDEL_BV_001: CTN must exist and be accessible
- - DOCDEL_BV_002: Document must exist and belong to CTN
- - DOCDEL_BV_003: User must have permission to delete documents
- - DOCDEL_BV_004: Cannot delete documents after visa submission
- - DOCDEL_BV_005: Cannot delete required documents unless replacing

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	success	Boolean	n1	M	Deletion success indicator (1=true, 0=false)	true

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
2	documentId	String (UUID)	an..50	M	Deleted document identifier, UUID format	f47ac10b-58cc-4372-a567-0e02b2c3d479
3	documentName	String	an..256	M	Name of deleted document, max 256 characters	Bill of Lading Original
4	deletedAt	DateTime	DATE	M	Deletion timestamp, ISO 8601 format, system generated	2025-11-14T16:45:00Z
5	deletedBy	Integer	n..10	M	User identifier who deleted document, positive integer	13345
6	message	String	an..200	M	Confirmation message, max 200 characters	Document successfully deleted

Error Codes:

- DOCDEL_E001: "CTN not found" (HTTP 404)
- DOCDEL_E002: "Document not found" (HTTP 404)
- DOCDEL_E003: "Cannot delete after visa submission" (HTTP 409)
- DOCDEL_E004: "Cannot delete required document" (HTTP 409)
- DOCDEL_E005: "Insufficient permissions" (HTTP 403)

Sample HTTP Request:

```
DELETE /api/ctns/503808/documents/f47ac10b-58cc-4372-a567-0e02b2c3d479
Content-Type: application/json
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
```

```
{
  "reason": "Incorrect document uploaded, need to replace with updated version"
}
```

Sample JSON Response:


```
{
  "success": true,
  "documentId": "f47ac10b-58cc-4372-a567-0e02b2c3d479",
  "documentName": "Bill of Lading Original",
  "deletedAt": "2025-11-14T16:45:00Z",
  "deletedBy": 13345,
  "message": "Document successfully deleted"
}
```

5.2.5 NIF Validation API (Angola Tax Registration Number)

Business Purpose: Validates NIF (Número de Identificação Fiscal) - Angola Tax Registration Number for importers, exporters, and consignees. Critical for ensuring tax compliance, preventing fraud, validating business entities, and meeting ARCCLA regulatory requirements for CNCA certificate processing.

Endpoint: POST /api/validation/nif

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	nifNumber	String	an..14	M	Angola Tax Registration Number (9-14 alphanumeric)	5000000000
2	entityName	String	an..200	O	Name of entity for cross-validation (max 200 chars)	ABC Trading Company
3	entityType	String	an..20	O	Type of entity (IMPORTER/EXPORTER/CONSIGNEE)	IMPORTER
4	validationType	String	an..20	O	Validation type (STANDARD/ENHANCED)	STANDARD

UI Validation Rules:

- NIF_UI_001: NIF number mandatory, 9-14 alphanumeric characters
- NIF_UI_002: Entity name optional, max 200 characters
- NIF_UI_003: Entity type from predefined list only
- NIF_UI_004: Remove leading/trailing spaces from NIF

Business Validation Rules:

- - NIF_BV_001: NIF must follow Angola tax authority format
- - NIF_BV_002: NIF must be active and valid in ARCCLA system
- - NIF_BV_003: Entity name must match registered name if provided
- - NIF_BV_004: Suspended or blacklisted NIFs cannot be used
- - NIF_BV_005: NIF entity type must match certificate type (import/export)

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	validation Status	String	an..20	M	Validation result (VALID/INVALID/SUSPENDED)	VALID
2	nifNumber	String	an..14	M	Validated NIF number (as submitted)	5000000000
3	registered Name	String	an..200	O	Official registered name from tax authority	ABC Trading Company Lda
4	entityStatus	String	an..20	O	Entity status (ACTIVE/INACTIVE/SUSPENDED)	ACTIVE
5	registrationDate	Date	DATE	O	NIF registration date (YYYY-MM-DD format)	2020-05-15
6	validation Code	String	an..10	M	Validation result code (system defined)	NIF_VALID
7	validation Message	String	an..500	M	Detailed validation message (max 500 chars)	NIF is valid and active
8	errorDetails	Array	-	O	Array of validation errors if any (empty on success)	[]

Error Codes:

- NIF_E001: "Invalid NIF format" (HTTP 400)
- NIF_E002: "NIF not found in ARCCLA system" (HTTP 404)
- NIF_E003: "NIF suspended or blacklisted" (HTTP 403)
- NIF_E004: "Entity name mismatch" (HTTP 400)

- NIF_E005: “NIF validation service unavailable” (HTTP 503)
- NIF_E006: “Invalid entity type for certificate” (HTTP 400)

Sample HTTP Request:

```
POST /api/validation/nif
Content-Type: application/json
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...

{
  "nifNumber": "5000000000",
  "entityName": "ABC Trading Company",
  "entityType": "IMPORTER",
  "validationType": "STANDARD"
}
```

Sample JSON Response (Success):

```
{
  "validationStatus": "VALID",
  "nifNumber": "5000000000",
  "registeredName": "ABC Trading Company Lda",
  "entityStatus": "ACTIVE",
  "registrationDate": "2020-05-15",
  "validationCode": "NIF_VALID",
  "validationMessage": "NIF is valid and active in ARCCLA system",
  "errorDetails": []
}
```

Sample JSON Response (Error):

```
{
  "validationStatus": "INVALID",
  "nifNumber": "5000000000",
  "registeredName": null,
  "entityStatus": null,
  "registrationDate": null,
  "validationCode": "NIF_NOT_FOUND",
  "validationMessage": "NIF not found in ARCCLA tax registration system",
  "errorDetails": [
    {
      "field": "nifNumber",
      "code": "NIF_E002",
      "message": "The provided NIF does not exist in the Angola tax authority database"
    }
  ]
}
```

Integration Notes: - NIF validation should be performed before CTN certificate creation -
 Cached validation results valid for 24 hours - Real-time validation with ARCCLA tax authority system - Supports bulk NIF validation (future enhancement)

5.2.6 Invoice Download API

Important Note on Invoice Applicability: > **CRITICAL:** Invoices are ONLY applicable for IMPORT profiles. Export profiles do NOT require invoice generation or download functionality. The invoice download API should only be called for import certificate types.

Business Purpose: Enables download of CNCA certificate invoices after certificate completion for import profiles only. Critical for financial record-keeping, customs documentation, and audit trail maintenance. **Not applicable for export certificates.**

Endpoint: GET /api/invoices/{invoiceId}/download

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	invoiceId	String	an..50	M	Unique invoice identifier (alphanumeric with hyphens)	INV-2024-001234
2	format	String	an..10	O	Invoice format (PDF/XML, default PDF)	PDF
3	certificate Type	String	an..20	O	Certificate type for validation (IMPORT/EXPORT)	IMPORT

Path Parameters: - invoiceId: Unique invoice identifier (mandatory)

Query Parameters: - format: Desired file format (optional, defaults to PDF)

UI Validation Rules:

- INV_UI_001: Invoice ID mandatory, alphanumeric with hyphens
- INV_UI_002: Format must be PDF or XML only
- INV_UI_003: Download only available for import profiles
- INV_UI_004: Display warning for export certificate requests

Business Validation Rules:

- INV_BV_001: Invoice must exist and be associated with authenticated user
- INV_BV_002: Certificate must be in “Approved” or “Issued” status

- - INV_BV_003: User must have permission to access invoice
- - INV_BV_004: Invoice download only allowed for IMPORT certificates
- - INV_BV_005: Export certificates should not generate or download invoices

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	fileName	String	an..100	M	Generated file name (format: invoice_{ID}.{ext})	invoice_INV-2024-001234.pdf
2	fileSize	Number	n..10	M	File size in bytes (positive integer)	245678
3	contentType	String	an..50	M	MIME type of file (application/pdf or text/xml)	application/pdf
4	downloadUrl	String	an..500	O	Direct download URL (if applicable, nullable)	https://j ul.adports.ae/downloads/...
5	invoiceData	Binary	-	M	Invoice file binary content (Base64 encoded)	[binary data]
6	generatedDate	Date	DATE	M	Invoice generation timestamp (ISO 8601 format)	2025-11-14T10:30:00Z

Error Codes:

- INV_E001: "Invoice not found" (HTTP 404)
- INV_E002: "Certificate not completed" (HTTP 400)
- INV_E003: "Access denied" (HTTP 403)
- INV_E004: "Invalid invoice format requested" (HTTP 400)
- INV_E005: "Export certificates do not support invoices" (HTTP 400)
- INV_E006: "Invoice generation in progress" (HTTP 202)

Sample API Request:

```
GET /api/invoices/INV-2024-001234/download?format=PDF
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
{
  "fileName": "invoice_INV-2024-001234.pdf",
  "fileSize": 245678,
  "contentType": "application/pdf",
  "downloadUrl": "https://jul.adports.ae/api/files/download/abc123xyz",
  "generatedDate": "2025-11-14T10:30:00Z",
  "invoiceData": "JVBERi0xLjQKJeLjz9MKMSAwIG9iago8PC9UeXB1L0NhdGFsb2cvUGFnZXMGMiAwIFI+..."
}
```

Invoice Applicability Rules:

Certificate Type	Invoice Required	Invoice Download Available	Notes
IMPORT	✅ Yes	✅ Yes	Invoice generated after approval
EXPORT	❌ No	❌ No	Export certificates do not require invoices

Important Implementation Notes:

- Pre-Download Validation:** Always validate the certificate type before allowing invoice download
- UI Display:** Do not show the invoice download button/option for export certificates
- Error Handling:** Return a specific error (INV_E005) when exporting the certificate attempts invoice download
- Business Logic:** Invoice generation workflow should only trigger for imported certificates
- Audit Trail:** Log all invoice download attempts with certificate type information

Integration Workflow:

- Certificate approved by ARCCLA
- For IMPORT certificates: Invoice generated automatically
- Certificate becomes available for download

4. User downloads invoice via API or UI
5. For EXPORT certificates: Skip invoice generation entirely

5.4 CTN Related Entity APIs

CTN Related Entity APIs manage the specific entities and components that are directly associated with CTN certificate processing, including consignee information, attachment management, and tracking data.

5.4.1 Consignees API

Business Purpose: Manages consignee information essential for CTN certificate processing and customs compliance. Critical for identifying cargo recipients, validating company details, ensuring proper documentation, and enabling accurate certificate issuance with verified consignee data.

Endpoint: GET /api/Consignees

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	\$sort	String	an..100	O	OData sort parameter for result ordering	NIFNumber
2	NIFNumber	String	an..50	O	Filter by NIF number (9-14 alphanumeric)	123456789
3	Company Name	String	an..200	O	Filter by company name (partial match, min 3 chars)	ACME
4	active	Boolean	n1	O	Filter active records only (1=active, 0=inactive)	1

UI Validation Rules:

- - CONSIGNEE_UI_001: NIF number format validation (9-14 digits)
- - CONSIGNEE_UI_002: Company name minimum 3 characters
- - CONSIGNEE_UI_003: Display with address for identification

Business Validation Rules:

- - CONSIGNEE_BV_001: NIF number must be unique and valid
- - CONSIGNEE_BV_002: Company must be registered and active
- - CONSIGNEE_BV_003: Consignee must have valid import license

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	Id	Integer	n..10	M	Unique consignee identifier, system generated	1
2	NIFNumber	String	an..50	M	Tax identification number (9-14 alphanumeric)	123456789
3	Company Name	String	an..200	M	Company/consignee name (max 200 chars)	ACME Import Company
4	ContactEmail	String	an..100	O	Contact email address (valid email format)	contact@acme.ao
5	Address	String	an..500	O	Company address (max 500 chars)	Rua da Independência 123
6	Active	Boolean	n1	M	Active status indicator (true/false)	true

Error Codes:

- CONSIGNEE_E001: "Invalid NIF number format" (HTTP 400)
- CONSIGNEE_E002: "Consignee not found" (HTTP 404)
- CONSIGNEE_E003: "Consignee not authorized for import" (HTTP 422)

Sample JSON Request:

```
GET /api/Consignees?${sort}=NIFNumber&active=1
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 1,
```



```

    "NIFNumber": "123456789",
    "CompanyName": "ACME Import Company",
    "ContactEmail": "contact@acme.ao",
    "Address": "Rua da Independencia 123, Luanda",
    "Active": true,
    "CreatedOn": "2023-01-15T10:30:00Z",
    "CreatedById": 1,
    "ModifiedOn": null,
    "ModifiedById": null
  }
]

```

5.4.3 CTN Tracking API

Business Purpose: Manages CTN shipment tracking information essential for logistics monitoring and cargo visibility. Critical for providing real-time shipment status, managing transport schedules, and enabling proactive logistics coordination throughout the shipping lifecycle.

Endpoint: GET /api/ctnTracking

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	\$expand	String	an..500	O	OData expand parameter (comma-separated entities)	CTN, DestinationPort, SourcePort, TransportType
2	ctn	Integer	n..10	M	CTN identifier (positive integer, required)	503808
3	ctnid	Integer	n..10	M	CTN identifier alternative (positive integer)	503808

UI Validation Rules:

- TRACKING_UI_001: CTN ID required for tracking lookup
- TRACKING_UI_002: Display tracking timeline visualization
- TRACKING_UI_003: Show estimated vs actual times with status

Business Validation Rules:

- TRACKING_BV_001: CTN must exist and be active

- TRACKING_BV_002: Tracking dates must be logical sequence
- TRACKING_BV_003: Vessel and voyage information must be consistent

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	Id	Integer	n..10	M	Tracking record identifier, system generated	71898
2	CTNId	Integer	n..10	M	Related CTN identifier (foreign key)	503808
3	SourcePortId	Integer	n..10	O	Source port identifier (foreign key, nullable)	150
4	DestinationPortId	Integer	n..10	O	Destination port identifier (foreign key, nullable)	275
5	TransportTypeId	Integer	n..10	M	Transport type identifier (foreign key, required)	1
6	ETD	DateTime	DATE	O	Estimated time of departure (ISO 8601, nullable)	2023-12-15T10:00:00Z
7	ETA	DateTime	DATE	O	Estimated time of arrival (ISO 8601, nullable)	2023-12-20T15:30:00Z
8	ATD	DateTime	DATE	O	Actual time of departure (ISO 8601, nullable)	2023-12-15T11:15:00Z
9	ATA	DateTime	DATE	O	Actual time of arrival (ISO 8601, nullable)	2023-12-20T14:45:00Z
10	VesselId	Integer	n..10	O	Vessel identifier (foreign key, nullable)	5163
11	VoyageNumber	String	an..50	O	Voyage number (alphanumeric, max 50 chars)	546894

Error Codes:

- TRACKING_E001: “Invalid CTN ID” (HTTP 400)
- TRACKING_E002: “Tracking information not found” (HTTP 404)
- TRACKING_E003: “Tracking data inconsistent” (HTTP 422)

Sample JSON Request:

```
GET /api/ctnTracking?$expand=CTN,DestinationPort,SourcePort,TransportType&ctn
id=503808
```

```
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
```

```
Accept: application/json
```

Sample JSON Response:

```
[
  {
    "Id": 71898,
    "CTNId": 503808,
    "SourcePortId": 80,
    "DestinationPortId": 35,
    "TransportTypeId": 1,
    "ETD": "2025-11-12T00:00:00.000Z",
    "ETA": "2025-12-09T00:00:00.000Z",
    "ATD": null,
    "ATA": null,
    "VesselId": 5163,
    "VoyageNumber": "546894",
    "CreatedOn": "2025-11-12T18:49:00Z",
    "CreatedBy": 13345,
    "ModifiedOn": null,
    "ModifiedBy": null
  }
]
```

5.3 File Management APIs

5.3.1 File Upload API

Endpoint: POST /api/fileupload

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	file	Binary	-	M	File upload (multipart/form-data, max 10MB)	[binary file data]

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
2	fileName	String	an..256	M	Original file name (alphanumeric with extensions)	Bill_of_Lading.pdf
3	description	String	an..500	O	File description (max 500 chars, optional)	Original shipping document

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	fileId	String	an..50	M	Unique file identifier (UUID format, system generated)	0da7461c-2155-4d20-a695-6b7463367327
2	fileName	String	an..256	M	Uploaded file name (as submitted)	Bill_of_Lading.pdf
3	fileSize	Integer	n..10	M	File size in bytes (positive integer)	245760
4	mimeType	String	an..100	M	File MIME type (auto-detected from file)	application/pdf
5	uploadDate	DateTime	DATE	M	Upload timestamp (ISO 8601 format, system generated)	2023-11-12T14:30:00Z

UI Validation Rules:

- **FILEUP_UI_001:** File size must not exceed 10MB
- **FILEUP_UI_002:** Only approved file types allowed (PDF, JPG, PNG, DOC, DOCX)
- **FILEUP_UI_003:** File name must not contain special characters

Business Validation Rules:

- **FILEUP_BV_001:** File must not be corrupted or malicious
- **FILEUP_BV_002:** Duplicate files should be detected and warned
- **FILEUP_BV_003:** User must have permission to upload files

Error Codes:

- **FILEUP_E001:** "File size exceeds maximum limit" (HTTP 413)
- **FILEUP_E002:** "File type not supported" (HTTP 415)
- **FILEUP_E003:** "File upload failed" (HTTP 500)

Sample Multipart Request:

```
POST /api/fileupload
Content-Type: multipart/form-data; boundary=----WebKitFormBoundary7MA4YWxkTrZu0gW
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...

-----WebKitFormBoundary7MA4YWxkTrZu0gW
Content-Disposition: form-data; name="file"; filename="Bill_of_Lading.pdf"
Content-Type: application/pdf

[Binary PDF file content]
-----WebKitFormBoundary7MA4YWxkTrZu0gW
Content-Disposition: form-data; name="fileName"

Bill_of_Lading.pdf
-----WebKitFormBoundary7MA4YWxkTrZu0gW
Content-Disposition: form-data; name="description"
```

Sample JSON Response:

```
{
  "fileId": "0da7461c-2155-4d20-a695-6b7463367327",
  "fileName": "Bill_of_Lading.pdf",
  "fileSize": 245760,
  "mimeType": "application/pdf",
  "uploadDate": "2023-11-12T14:30:00Z"
}
```

GET /api/filedownload/{fileId} - Download File

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	fileId	String	an..50	M	File identifier from URL path (UUID format)	0da7461c-2155-4d20-a695-6b7463367327

Sample Download Request:

```
GET /api/filedownload/0da7461c-2155-4d20-a695-6b7463367327
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	contentStream	Binary	-	M	Base64 encoded file content	JVBERi0xLjQKJeLjz9MK...
2	fileName	String	an..256	M	Original file name	Bill_of_Lading.pdf
3	mimeType	String	an..100	M	File MIME type	application/pdf
4	fileSize	Integer	n..10	M	File size in bytes	245760

Sample Download Response:

```
{
  "contentStream": "JVBERi0xLjQKJeLjz9MKNSAwIG9iago8PC9MZWN5dGggNzggL0ZpbHRlc",
  "fileName": "Bill_of_Lading.pdf",
  "mimeType": "application/pdf",
  "fileSize": 245760
}
```

5.6 Specialized CTN APIs

5.6.1 Parent CTN Relationships API

Endpoint: GET /api/Ctns/GetAllowedParentCtns

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	ctnId	Integer	n..10	M	Current CTN identifier (positive integer, required)	503808

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
2	\$sort	String	an..100	O	OData sort parameter for result ordering	BL_number
3	BL_number	String	an..50	O	Filter by Bill of Lading number (partial match)	MSC123456

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	Id	Integer	n..10	M	Parent CTN ID (system generated identifier)	503800
2	CTN_Reference_Number	String	an..50	M	Parent CTN reference (format: AO-CNT-{ID}-{YEAR})	AO-CNT-503800-2023
3	BL_number	String	an..50	M	Parent Bill of Lading (alphanumeric)	MSC123456
4	StatusId	Integer	n..10	M	Parent CTN status (system defined status code)	2
5	CanBeParent	Boolean	n1	M	Eligibility as parent (true/false, dynamically calculated)	true

5.6.2 CTN Export API

Endpoint: GET /api/ctns/export

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	\$expand	String	an..500	O	OData expand parameter (comma-separated entity names)	CargoType,Status,Visum_Agent

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
2	\$export_format	String	an..10	M	Export format (csv/xlsx/json, required)	csv
3	\$sort	String	an..100	O	OData sort parameter for result ordering	- ModifiedOn
4	\$top	Integer	n..5	O	Number of records to export (max 1000, default all)	10
5	\$filter	String	an..1000	O	OData filter expression for data selection	StatusId eq 2

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	Content	Text/CSV	-	M	CSV formatted CTN data (or xlsx/json based on format parameter)	"Id","CTN_ReferenceNumber".
2	ContentType	String	an..50	M	Response content type (text/csv or application/vnd.ms-excel or application/json)	text/csv
3	FileName	String	an..100	M	Suggested file name (format: CTN_Export_{YYYYMMDD}.{ext})	CTN_Export_20231112.csv

6. Enhanced API Capabilities

6.1 Amendment APIs

Amendment APIs provide controlled modification capabilities for existing certificates with ARCCLA approval workflow.

6.1.1 Amendment Request Creation API

Endpoint: POST /api/CTNs/{id}/amendments

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	id	Integer	n..10	M	CTN identifier from URL path (positive integer)	503808
2	requestReason	String	an..500	M	Reason for amendment request (max 500 chars)	Incorrect vessel name provided
3	businessJustification	String	an..1000	O	Detailed business justification (max 1000 chars)	Customer informed of vessel change
4	requestedChanges	Array	-	M	Array of field changes requested (at least 1 change)	[field change objects]
5	urgencyLevel	String	an..20	O	Urgency level (Normal/High/Critical, default Normal)	Normal

Requested Changes Array Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	fieldPath	String	an..100	M	JSON path to field being changed (dot notation)	VesselName
2	fieldName	String	an..100	M	Human-readable field name (max 100 chars)	Vessel Name
3	oldValue	String	an..500	M	Current field value (max 500 chars)	MSC MEDITERRANEAN
4	newValue	String	an..500	M	Proposed new field value (max 500 chars)	MAERSK ESSEX

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
5	changeReason	String	an..200	M	Specific reason for this field change (max 200 chars)	Vessel change notification

Sample JSON Request:

```
{
  "requestReason": "Incorrect vessel name provided",
  "businessJustification": "Customer informed of vessel change after booking confirmation",
  "urgencyLevel": "Normal",
  "requestedChanges": [
    {
      "fieldPath": "VesselName",
      "fieldName": "Vessel Name",
      "oldValue": "MSC MEDITERRANEAN",
      "newValue": "MAERSK ESSEX",
      "changeReason": "Vessel change notification from carrier"
    },
    {
      "fieldPath": "ETA",
      "fieldName": "Estimated Time of Arrival",
      "oldValue": "2023-12-20T15:30:00Z",
      "newValue": "2023-12-22T10:00:00Z",
      "changeReason": "Updated ETA due to vessel change"
    }
  ]
}
```

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	amendmentId	String	an..50	M	Unique amendment identifier (UUID format, system generated)	a1b2c3d4-e5f6-7890-abcd-1234567890ef
2	amendmentNumber	String	an..50	M	Sequential amendment number (format: AMD-{CTN}-{SEQ})	AMD-503808-001

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
3	status	String	an..20	M	Amendment status (Pending/UnderReview/Approved/Rejected)	Pending
4	requestDate	DateTime	DATE	M	Amendment request timestamp (ISO 8601, system generated)	2025-11-13T10:45:00Z
5	estimatedFee	Decimal	n..10,2	O	Estimated amendment fee in USD (max 2 decimals)	25.00

Sample JSON Response:

```
{
  "amendmentId": "a1b2c3d4-e5f6-7890-abcd-1234567890ef",
  "amendmentNumber": "AMD-503808-001",
  "originalCertificateId": "503808",
  "status": "Pending",
  "requestDate": "2025-11-13T10:45:00Z",
  "estimatedFee": 25.00,
  "currency": "USD",
  "submittedBy": 13345,
  "submittedByRole": "CustomsBroker"
}
```

6.1.2 Amendment Status API

Endpoint: GET /api/amendments/{amendmentId}/status

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	amendmentId	String	an..50	M	Amendment identifier from URL path (UUID format)	a1b2c3d4-e5f6-7890-abcd-1234567890ef

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	amendmentId	String	an..50	M	Amendment identifier (UUID format, as requested)	a1b2c3d4-e5f6-7890-abcd-1234567890ef
2	amendmentNumber	String	an..50	M	Amendment number (format: AMD-{CTN}-{SEQ}, system generated)	AMD-503808-001
3	status	String	an..20	M	Current amendment status (from predefined status enum)	UnderReview
4	assignedTo	String	an..100	O	ARCCLA broker assigned (email format, nullable if unassigned)	John.Smith@arccla.ao
5	lastUpdated	DateTime	DATE	M	Last status update timestamp (ISO 8601 format)	2025-11-13T14:30:00Z
6	statusHistory	Array	-	O	Amendment status history (array of status change objects)	[...]

6.2 Cancellation APIs

Cancellation APIs provide controlled termination of certificate requests with financial impact handling.

6.2.1 Cancellation Request Creation API

Endpoint: POST /api/CTNs/{id}/cancellations

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	id	Integer	n..10	M	CTN identifier from URL path (positive integer)	503808
2	reasonCode	String	an..20	M	Standardized cancellation reason code (predefined codes)	SHIPMENT_CANCELLED
3	reasonDescription	String	an..500	M	Detailed cancellation reason (max 500 chars)	Shipment cancelled by customer
4	additionalDetails	String	an..1000	O	Additional cancellation details (max 1000 chars)	Customer business closure
5	urgencyLevel	String	an..20	O	Cancellation urgency (Normal/High/Critical, default Normal)	High

Sample JSON Request:

```
{
  "reasonCode": "SHIPMENT_CANCELLED",
  "reasonDescription": "Shipment cancelled by customer due to business closure",
  "additionalDetails": "Customer has permanently ceased operations and requires immediate cancellation",
  "urgencyLevel": "High"
}
```

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	cancellationId	String	an..50	M	Unique cancellation identifier (UUID format, system generated)	x1y2z3a4-b5c6-7890-defg-9876543210ab

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
2	cancellationNumber	String	an..50	M	Sequential cancellation number (format: CAN-{CTN}-{SEQ})	CAN-503808-001
3	status	String	an..20	M	Cancellation status (Pending/UnderReview/Approved/Rejected)	Pending
4	requestDate	DateTime	DATE	M	Cancellation request timestamp (ISO 8601, system generated)	2025-11-13T11:30:00Z
5	estimatedRefund	Decimal	n..10,2	O	Estimated refund amount in USD (max 2 decimals)	125.00

Sample JSON Response:

```
{
  "cancellationId": "x1y2z3a4-b5c6-7890-defg-9876543210ab",
  "cancellationNumber": "CAN-503808-001",
  "originalCertificateId": "503808",
  "status": "Pending",
  "requestDate": "2025-11-13T11:30:00Z",
  "estimatedRefund": 125.00,
  "refundCurrency": "USD",
  "submittedBy": 13345,
  "submittedByRole": "CustomsBroker"
}
```

6.2.2 Cancellation Status API

Endpoint: GET /api/cancellations/{cancellationId}/status

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	cancellationId	String	an..50	M	Cancellation identifier from URL path (UUID format)	x1y2z3a4-b5c6-7890-defg-9876543210ab

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	cancellationId	String	an..50	M	Cancellation identifier (UUID format, as requested)	x1y2z3a4-b5c6-7890-defg-9876543210ab
2	cancellationNumber	String	an..50	M	Cancellation number (format: CAN-{CTN}-{SEQ}, system generated)	CAN-503808-001
3	status	String	an..20	M	Current cancellation status (from predefined status enum)	UnderReview
4	assignedTo	String	an..100	O	ARCCLA broker assigned (email format, nullable if unassigned)	Maria.Santos@arccla.ao
5	assignedToName	String	an..200	O	Full name of assigned broker (for display purposes)	Maria Santos
6	refundStatus	String	an..20	O	Refund processing status (Pending/Processing/Completed/Rejected, nullable)	Pending
7	refundAmount	Decimal	n..10,2	O	Refund amount in USD (max 2 decimals, nullable if not calculated)	125.00
8	lastUpdated	DateTime	DATE	M	Last status update timestamp (ISO 8601 format)	2025-11-13T15:45:00Z
9	lastUpdatedBy	String	an..100	O	User who last updated status (username, nullable)	ARCCLA_Admin

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
10	originalCertificateId	String	an..50	O	Related certificate ID (CTN identifier)	503808
11	requestDate	DateTime	DATE	O	Cancellation request date (ISO 8601 format)	2025-11-13T11:30:00Z
12	statusHistory	Array	-	O	Status change history (array of status change objects)	[...]

6.4 Status Polling APIs

Status polling APIs provide real-time status updates with eligibility flags for allowed operations.

6.4.1 Status Retrieval API (Polling-Based Status Updates)

Business Purpose: Enables JUL system to retrieve the current status of CNCA certificates by polling the SINTECE system. Replaces webhook-based notifications with pull-based status updates, providing reliable status synchronization, eliminating callback URL management, and supporting firewall-friendly integration patterns.

Endpoint: GET /api/ctns/{ctnId}/status

Request Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	ctnId	Integer	n..10	M	CTN certificate identifier from path (positive integer)	503808
2	includeHistory	Boolean	n1	O	Include status change history (true/false, default false)	false
3	includeDetails	Boolean	n1	O	Include detailed status information (true/false, default true)	true

Path Parameters: - ctnId: CTN certificate identifier (mandatory)

Query Parameters: - includeHistory: Set to true to get full status change history -
includeDetails: Set to true to get detailed status information

UI Validation Rules:

- **STATUS_UI_001:** CTN ID mandatory, positive integer
- **STATUS_UI_002:** Boolean parameters only accept true/false
- **STATUS_UI_003:** Display loading state during status retrieval

Business Validation Rules:

- **STATUS_BV_001:** CTN must exist in system
- **STATUS_BV_002:** User must have permission to view CTN status
- **STATUS_BV_003:** Status must be current (not stale data)

Response Elements:

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	ctnId	Integer	n..10	M	CTN certificate identifier (as requested)	503808
2	ctnReferenceNumber	String	an..50	M	System generated CTN reference (format: {ID} or full)	170543
3	currentStatus	String	an..50	M	Current certificate status (from predefined status list)	Approved
4	statusId	Integer	n..10	M	Status identifier (system defined)	3
5	statusUpdatedAt	DateTime	DATE	M	Last status update timestamp (ISO 8601 format)	2025-11-14T10:30:00Z
6	statusUpdatedBy	String	an..100	O	User who updated status (username, nullable)	ARCCLA_Broker_123
7	canAmend	Boolean	n1	M	Amendment eligibility flag (true/false, dynamically calculated)	true
8	canCancel	Boolean	n1	M	Cancellation eligibility flag (true/false,	false

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
					dynamically calculated)	
9	statusMessage	String	an..500	O	Status description or notes (max 500 chars, nullable)	Certificate approved by ARCCLA
10	nextAction	String	an..200	O	Recommended next action (max 200 chars, nullable)	Process payment and download
11	statusHistory	Array	-	O	Status change history (array, only if includeHistory=true)	[...]

Status History Element (when includeHistory=true):

S. No	Attributes	Data Type	Format	MoE	Derivation Logic	Data Example
1	statusId	Integer	n..10	M	Status identifier (system defined)	2
2	statusName	String	an..50	M	Status name (from predefined status list)	Submitted
3	changedAt	DateTime	DATE	M	Status change timestamp (ISO 8601 format)	2025-11-13T14:20:00Z
4	changedBy	String	an..100	O	User who changed status (username, nullable)	Customs_Broker_456
5	remarks	String	an..500	O	Status change remarks (max 500 chars, nullable)	Submitted for ARCCLA review

Error Codes:

- **STATUS_E001:** "CTN not found" (HTTP 404)
- **STATUS_E002:** "Access denied" (HTTP 403)
- **STATUS_E003:** "Invalid CTN ID format" (HTTP 400)
- **STATUS_E004:** "Status service temporarily unavailable" (HTTP 503)

Sample API Request:

```
GET /api/ctns/503808/status?includeHistory=true&includeDetails=true
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Sample JSON Response:

```
{
  "ctnId": 503808,
  "ctnReferenceNumber": "170543",
  "currentStatus": "Approved",
  "statusId": 3,
  "statusUpdatedAt": "2025-11-14T10:30:00Z",
  "statusUpdatedBy": "ARCCLA_Broker_123",
  "canAmend": true,
  "canCancel": false,
  "statusMessage": "Certificate approved by ARCCLA. Ready for payment process
ing.",
  "nextAction": "Process payment and download certificate",
  "statusHistory": [
    {
      "statusId": 1,
      "statusName": "Created",
      "changedAt": "2025-11-12T18:49:30Z",
      "changedBy": "Customs_Broker_456",
      "remarks": "Initial CTN creation"
    },
    {
      "statusId": 2,
      "statusName": "Submitted",
      "changedAt": "2025-11-13T14:20:00Z",
      "changedBy": "Customs_Broker_456",
      "remarks": "Submitted for ARCCLA review"
    },
    {
      "statusId": 3,
      "statusName": "Approved",
      "changedAt": "2025-11-14T10:30:00Z",
      "changedBy": "ARCCLA_Broker_123",
      "remarks": "Certificate approved by ARCCLA"
    }
  ]
}
```

```

    ]
  }

```

Polling Strategy Recommendations:

Certificate Status	Recommended Polling Interval	Notes
Created	No polling needed	Local status
Submitted	Every 5 minutes	Active review period
UnderReview	Every 10 minutes	ARCCLA processing
PendingInfo	Every 30 minutes	Waiting for trader response
Approved	Every 2 minutes	Payment processing
Issued	No polling needed	Final status
Cancelled	No polling needed	Final status

Implementation Guidelines:

1. **Efficient Polling:** Use appropriate intervals based on certificate status
2. **Rate Limiting:** Respect API rate limits (max 120 requests/minute per user)
3. **Caching:** Cache status responses for 30 seconds to reduce API calls
4. **Error Handling:** Implement exponential backoff for failed polling attempts
5. **User Experience:** Show real-time updates in UI based on polling results
6. **Background Jobs:** Use scheduled jobs for polling instead of continuous loops
7. **Notification:** Trigger user notifications when status changes detected

Advantages Over Webhooks: -

- ✓ No callback URL configuration required
- ✓ Firewall-friendly (outbound requests only)
- ✓ Better control over update frequency
- ✓ Simpler security model (no webhook signature validation)
- ✓ Easier error handling and retry logic
- ✓ No missed notifications due to network issues

7. API Implementation Summary

6.1 Complete CNCA Certificate API Coverage

The JUL-SINTECE Integration Control Document provides comprehensive API documentation covering the complete CNCA certificate lifecycle with section-by-section submission capabilities.

API Coverage Statistics:

- **Total APIs Documented:** 31 comprehensive APIs

- **Master Data APIs:** 15 essential reference data endpoints
- **Certificate Management APIs:** 11 core certificate workflow endpoints
- **CTN Related Entity APIs:** 5 supporting data endpoints
- **JSON Coverage:** 100% of APIs include production-ready request/response samples

Section-by-Section Certificate Submission APIs:

API Category	Endpoint	Purpose	Production Data
Core Creation	POST /api/ctns	Create initial CTN record	✓ CTN 503808
Address Management	POST /api/ctnAddresses	Add shipper/consignee/forwarder details	✓ Live addresses
Cargo Details	POST /api/ctnGoods	Add goods classifications and values	✓ Cargo data
Container Info	POST /api/ctnContainers	Add container types and numbers	✓ Container specs
Transport Routes	POST /api/ctnTracking	Add vessel and port information	✓ Route data
Document Attachment	POST /api/ctnAttachments	Link supporting documents	✓ File links
Final Submission	POST /api/ctns/actions/requestvisa/{id}	Submit for approval	✓ Visa requests

Key Integration Benefits:

1. **Progressive Data Entry:** Section-by-section approach improves user experience and data quality
2. **Comprehensive Validation:** Field-level and business rule validation at each step
3. **Production Ready:** All APIs tested with real SINTECE system data
4. **Complete Workflow:** From initial creation to final certificate approval
5. **Error Handling:** Detailed error codes and validation messages
6. **Audit Trail:** Complete tracking of certificate submission process

 **Workflow Completeness:** -  **Initial Creation:** Basic CTN record with core information -  **Data Enhancement:** Section-wise addition of detailed information - 
Document Management: File upload and attachment linking -  **Validation & Submission:** Complete data validation before approval request -  **Status Tracking:** Real-time status monitoring throughout process -  **Amendment Support:** Post-issuance certificate modification capabilities -  **Cancellation Support:** Certificate cancellation workflow

6.2 Implementation Readiness

 **Development Team Benefits:** - Complete API specifications with production JSON samples - Comprehensive error code documentation - Business rule validation requirements - Real system response examples for testing - Step-by-step integration guidance

 **Business Process Coverage:** - End-to-end CNCA certificate lifecycle - Section-by-section submission for improved UX - Complete supporting document management - Automated validation and approval workflows - Real-time status tracking and notifications

8. Validation Framework

7.1 Field-Level Validation Rules

Text Field Validations:

- **TXT_001:** Required fields cannot be null or empty
- **TXT_002:** Text fields must not exceed maximum length
- **TXT_003:** Special characters properly escaped for security

Numeric Field Validations:

- **NUM_001:** Numeric fields must contain valid numbers
- **NUM_002:** Currency amounts non-negative with max 2 decimal places
- **NUM_003:** Integer IDs must be positive

Date/Time Validations:

- **DT_001:** All dates in ISO 8601 format (YYYY-MM-DDTHH:mm:ssZ)
- **DT_002:** ETD must be before or equal to ETA
- **DT_003:** Creation dates cannot be in future

7.2 Business Logic Validation

Trade Compliance:

- **TC_001:** Angola must be origin or destination for CNCA certificates
- **TC_002:** Sanctioned countries require special compliance screening
- **TC_003:** High-value shipments (>\$50,000) require enhanced documentation

Financial Validations:

- **FIN_001:** Invoice value must match bill of lading value ($\pm 5\%$ tolerance)
- **FIN_002:** Currency conversion rates current (within 24 hours)
- **FIN_003:** Payment terms consistent with Incoterms selected

Workflow State Validations:

- **ST_001:** Only valid status transitions allowed per business rules
- **ST_002:** Required actions completed before status advancement
- **ST_003:** Role-based permissions validated for status changes

7.3 Error Handling Standards

Error Response Structure:

```
{
  "error": {
    "code": "CTN_V001",
    "message": "Bill of Lading number must be unique",
    "severity": "ERROR",
    "field": "BL_number",
    "timestamp": "2025-11-13T16:45:00Z"
  }
}
```

Performance Standards: - **PERF_001:** Field validations complete within 100ms -
PERF_002: Business validations complete within 500ms - **PERF_003:** External validations complete within 2 seconds

8.3 Authentication and Authorization

The JUL-SINTECE integration uses Keycloak as the primary Single Sign-On (SSO) solution for centralized authentication and authorization management. Keycloak provides enterprise-grade identity and access management capabilities, ensuring secure and seamless authentication across both JUL and SINTECE systems.

8.3.1 SSO Solution - Keycloak

Overview: The JUL-SINTECE integration uses Keycloak as the primary Single Sign-On (SSO) solution for centralized authentication and authorization management. Keycloak provides enterprise-grade identity and access management capabilities, ensuring secure and seamless authentication across both JUL and SINTECE systems.

Key Features:

- Centralized user authentication and session management
- Single Sign-On (SSO) across multiple applications
- Token-based authentication with JWT
- Role-based access control (RBAC)

- Multi-factor authentication (MFA) support
- User federation and identity brokering

8.3.2 Supported Authentication Protocols

The client authentication system has the capability to integrate with Keycloak using the following protocols:

Protocol	Description	Use Case
OpenID Connect	Authentication layer built on OAuth 2.0	Recommended for user authentication and SSO
OAuth 2.0	Authorization framework for delegated access	API authorization and service-to-service communication
SAML 2.0	XML-based authentication protocol	Enterprise SSO integration (if required)

Primary Protocol: OpenID Connect (OIDC) is the recommended protocol for JUL-SINTECE integration.

8.3.3 Client Integration Capabilities

Current System Capabilities:

- JWT token support and validation
- Token-based authentication flow
- Support for Bearer token authentication
- Client credentials flow for service-to-service communication
- Authorization code flow for user authentication

Integration Approach:

1. **Client Registration:** Register JUL application in Keycloak realm
2. **Client Credentials:** Obtain client ID and client secret
3. **Token Endpoint:** Configure token endpoint for authentication
4. **User Info Endpoint:** Configure user information retrieval
5. **Logout Endpoint:** Implement proper logout and session management

8.3.4 JWT Token Management

Token Format and Structure: The system uses JSON Web Tokens (JWT) for secure authentication. All API requests must include a valid JWT token in the Authorization header.

Token Request:

```
POST /auth/realms/{realm}/protocol/openid-connect/token
Content-Type: application/x-www-form-urlencoded
```



```
grant_type=password&
client_id=jul-client&
client_secret={client_secret}&
username={username}&
password={password}
```

Token Response:

```
{
  "access_token": "eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...",
  "expires_in": 3600,
  "refresh_expires_in": 1800,
  "refresh_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "Bearer",
  "not-before-policy": 0,
  "session_state": "d2d2b3b4-4f5e-4f3e-9c1a-5f6e7d8e9f0a",
  "scope": "profile email"
}
```

Token Usage in API Requests: All API requests must include the access token in the Authorization header:

```
GET /api/ctns
Authorization: Bearer eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9...
Accept: application/json
```

Token Validation: - Tokens are validated on every API request - Signature verification using public key from Keycloak - Expiration time validation - Issuer and audience claims validation - Required scopes/roles validation

Token Refresh: When the access token expires, use the refresh token to obtain a new access token:

```
POST /auth/realms/{realm}/protocol/openid-connect/token
Content-Type: application/x-www-form-urlencoded

grant_type=refresh_token&
client_id=jul-client&
client_secret={client_secret}&
refresh_token={refresh_token}
```

Token Lifecycle:

- **Access Token Lifetime:** 60 minutes (configurable)
- **Refresh Token Lifetime:** 30 minutes (configurable)
- **Session Timeout:** Configurable per realm
- **Token Revocation:** Supported via logout endpoint

8.3.5 Integration Requirements

Technical Requirements:

1. **Keycloak Server:** Version 21.0 or higher
2. **TLS/SSL:** All communication must use HTTPS
3. **Network Access:** JUL system must have network access to Keycloak server
4. **Client Configuration:** Proper client registration in Keycloak
5. **Role Mapping:** Define and map user roles for authorization

Security Requirements:

- Secure storage of client credentials
- Token encryption in transit
- Token validation on every API call
- Proper error handling for authentication failures
- Session timeout and automatic logout

Implementation Checklist:

- [] Register JUL application as Keycloak client
- [] Configure redirect URLs and valid origins
- [] Set up user roles and permissions
- [] Implement token acquisition flow
- [] Implement token refresh mechanism
- [] Add authorization header to all API calls
- [] Implement proper logout functionality
- [] Test token expiration and refresh
- [] Implement error handling for auth failures
- [] Document authentication flows for developers